US 20250280915A1

(19) United States
(12) Patent Application Publication
Zeilinger et al.
(10) Pub. No.: US 2025/0280915 A1
(43) Pub. Date: Sep. 11, 2025

(54) **HARD HAT ATTACHMENT SYSTEM AND SUN VISOR**

(71) Applicant: **Milwaukee Electric Tool Corporation**, Brookfield, WI (US)

(72) Inventors: **Todd Andrew Zeilinger**, Wauwatosa, WI (US); **Patrick W. McCarthy**, Milwaukee, WI (US); **Caleb C. Adams**, Milwaukee, WI (US); **Nicole Z. Summersett**, Milwaukee, WI (US); **Scott Michael Cline**, Menomonee Falls, WI (US)

(21) Appl. No.: **19/217,009**

(22) Filed: **May 23, 2025**

Related U.S. Application Data

(63) Continuation of application No. 17/461,163, filed on Aug. 30, 2021, now Pat. No. 12,336,586, which is a continuation of application No. PCT/US2021/045405, filed on Aug. 10, 2021.

(60) Provisional application No. 63/066,561, filed on Aug. 17, 2020, provisional application No. 63/087,578, filed on Oct. 5, 2020, provisional application No. 63/162,736, filed on Mar. 18, 2021, provisional application No. 63/167,458, filed on Mar. 29, 2021.

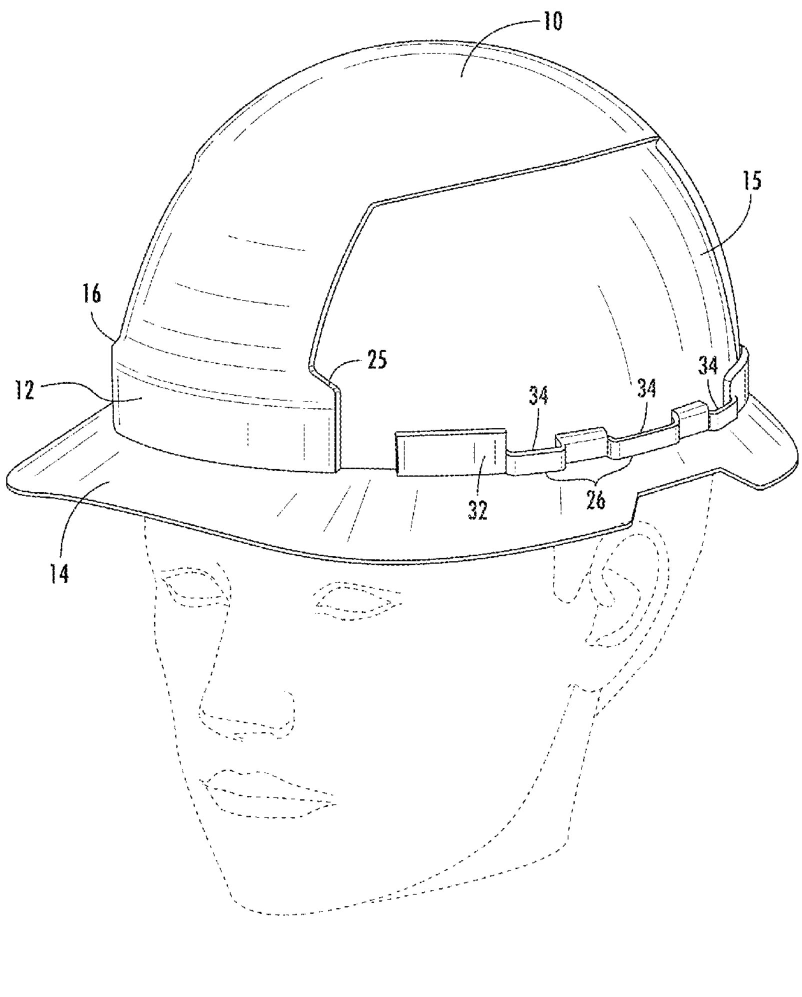
Publication Classification

(51) **Int. Cl.**
A42B 3/22 (2006.01)
A42B 3/04 (2006.01)
A42B 3/18 (2006.01)

(52) **U.S. Cl.**
CPC ***A42B 3/22*** (2013.01); ***A42B 3/0406*** (2013.01); ***A42B 3/185*** (2013.01)

(57) **ABSTRACT**

A universal mounting system for a hard hat is provided that includes front and rear mounting locations and auxiliary ridges on either side. A visor is coupled to the hard hat mounting system in such a way as to provide access with minimal interference from the visor to the equipment attached to the mounting locations and ports in the auxiliary ridges. Various embodiments of visors enable modular designs that incorporate different materials and features of the visor. Parallel stretch zones enable folding and holding opposite brims of the visor. Enclosed elastic areas along the inner sides of visor provide user access to auxiliary ridges. Combinations of various accessories on hard hat and how the accessories work together on the universal mounting system are discussed: such as the interaction between visors, earmuffs, face shields, and lamps.



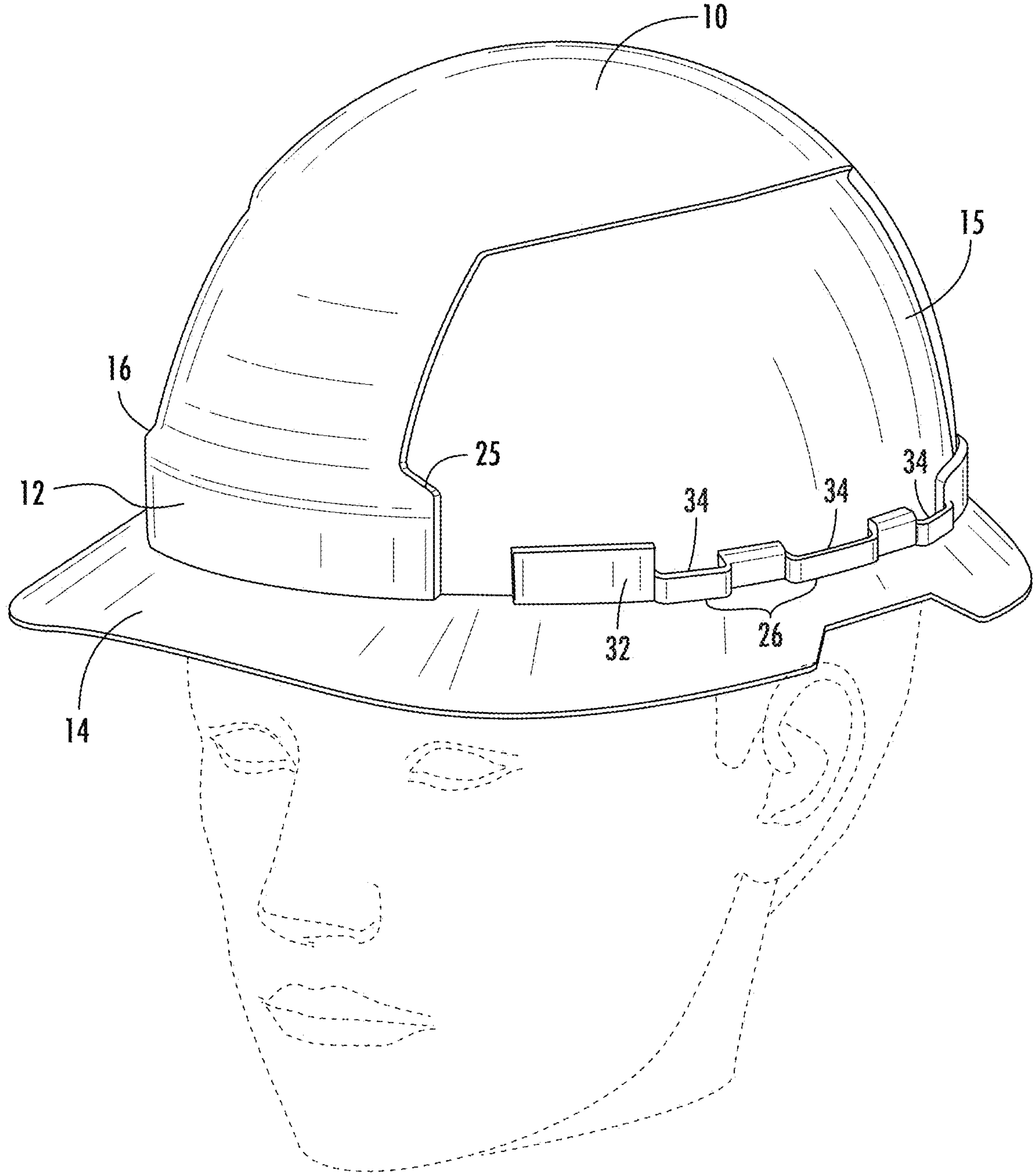


FIG. 1

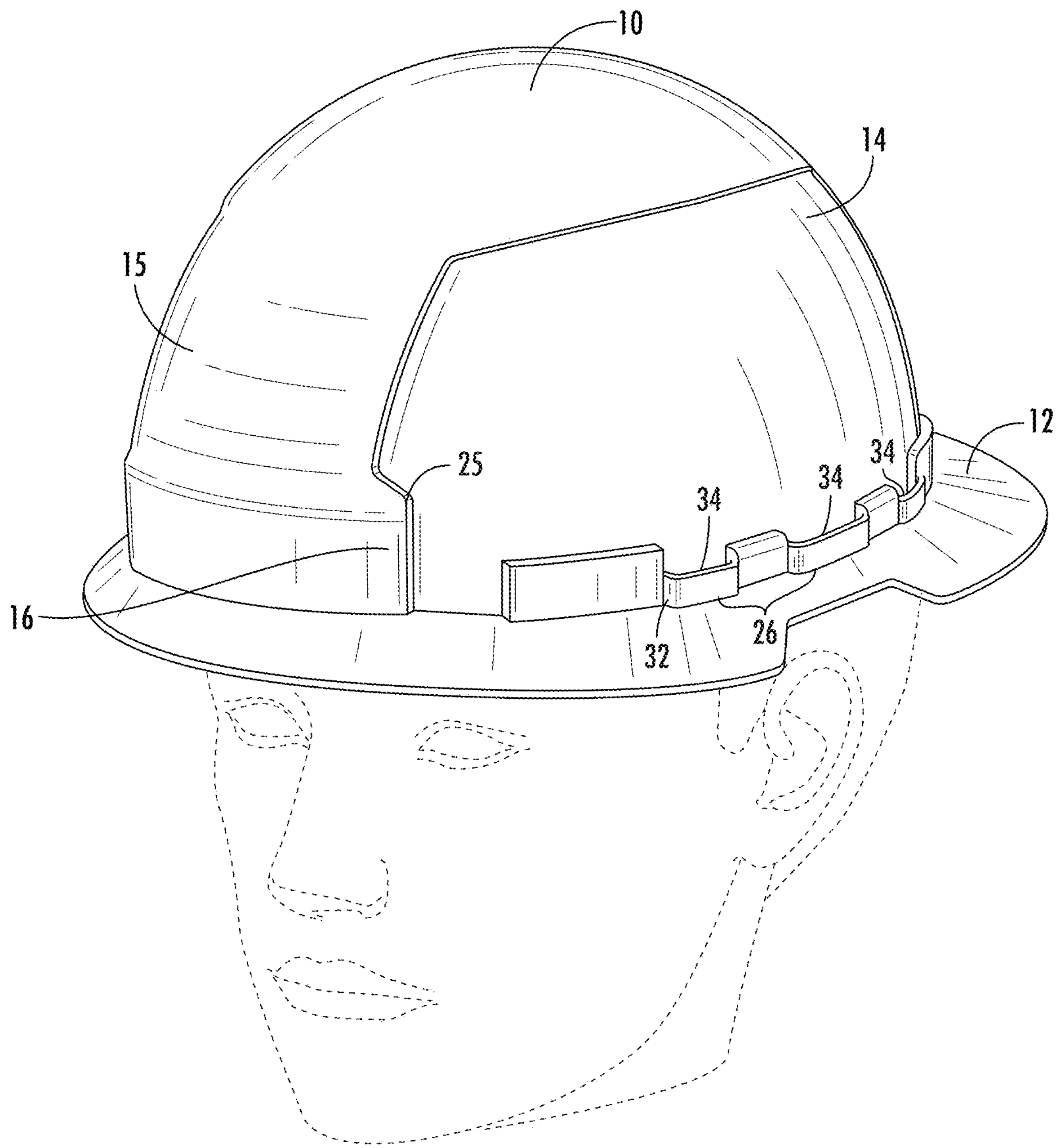


FIG. 2

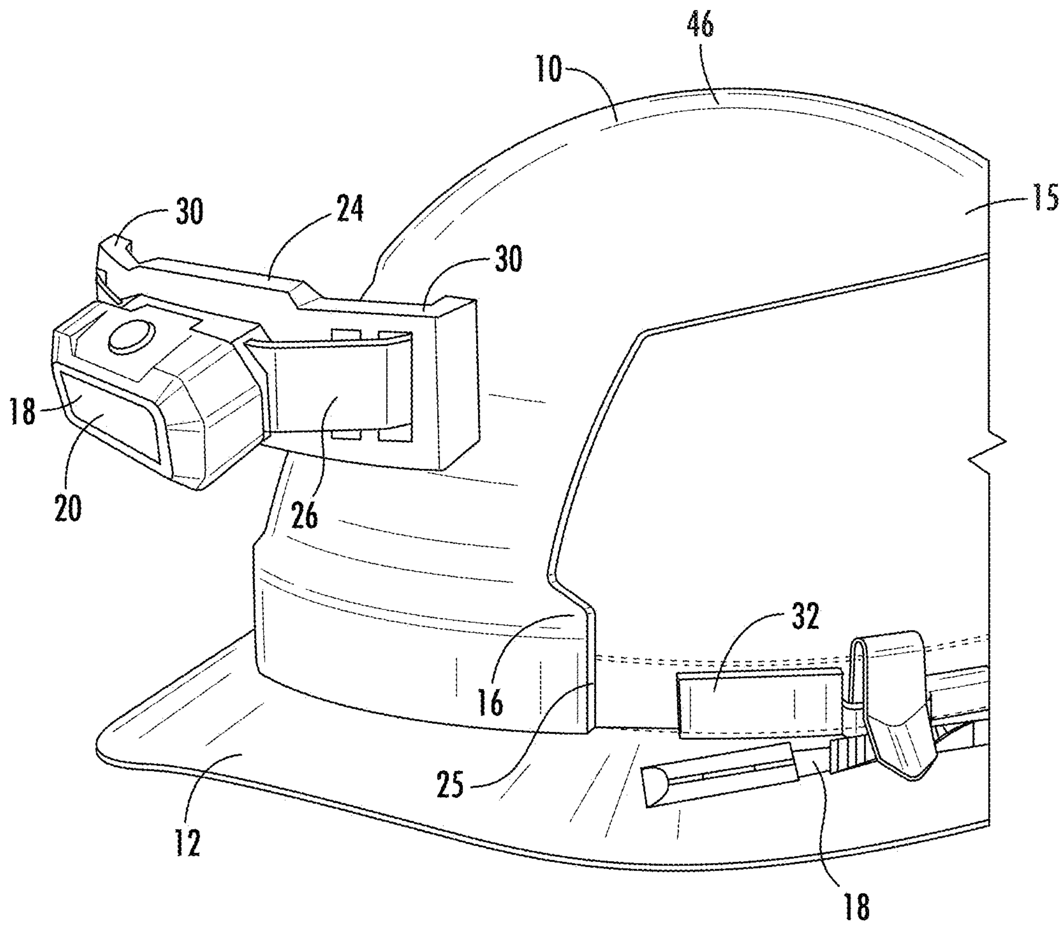


FIG. 3

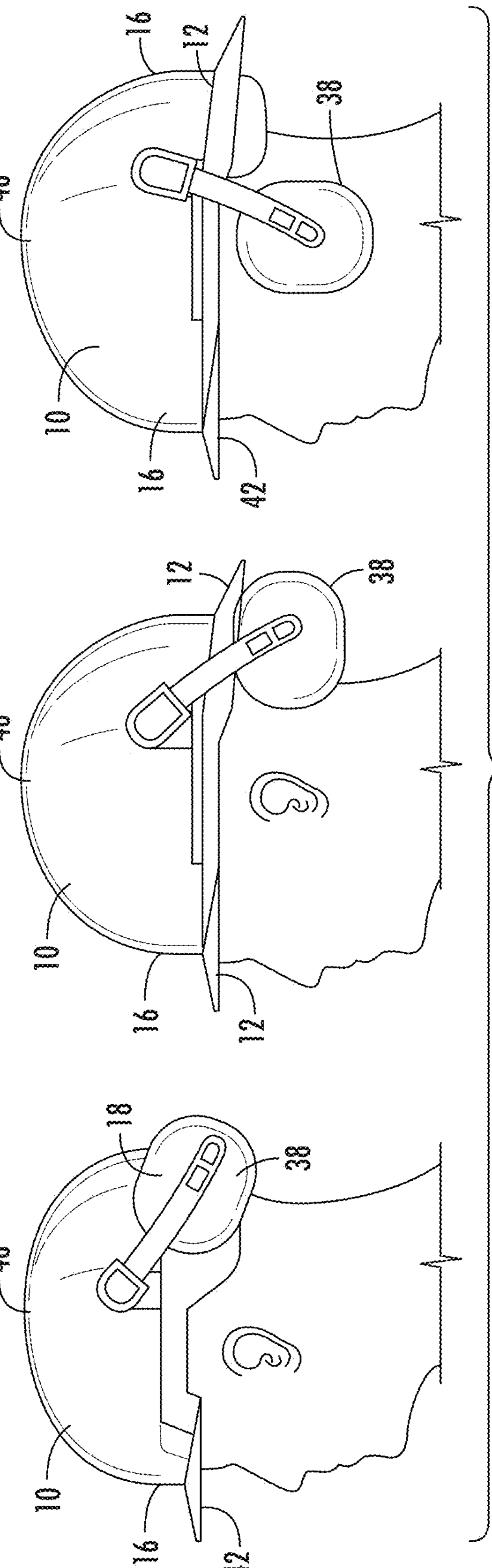


FIG. 4

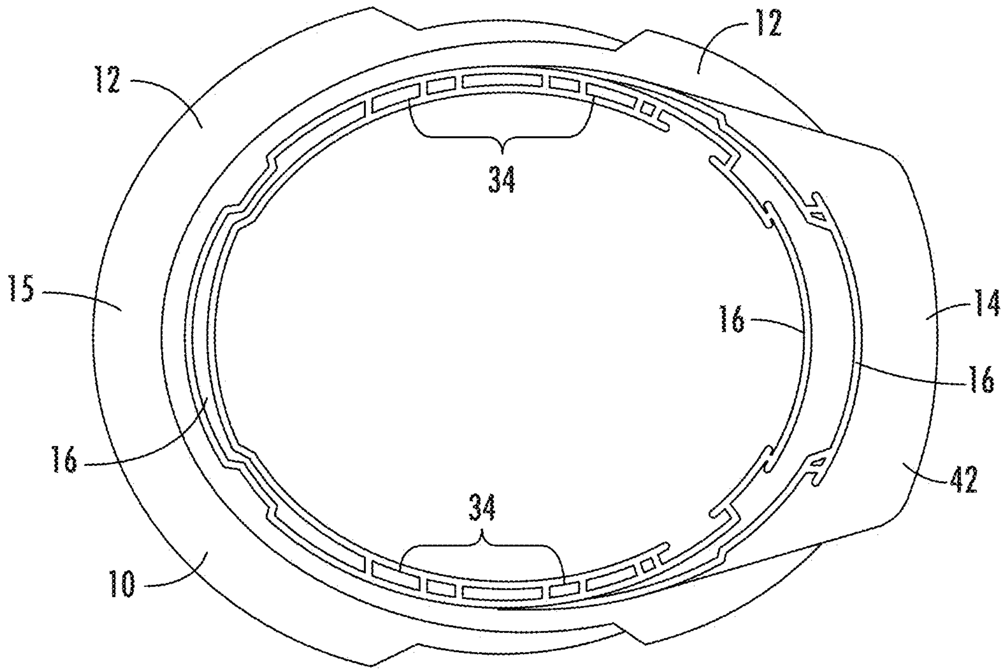


FIG. 5

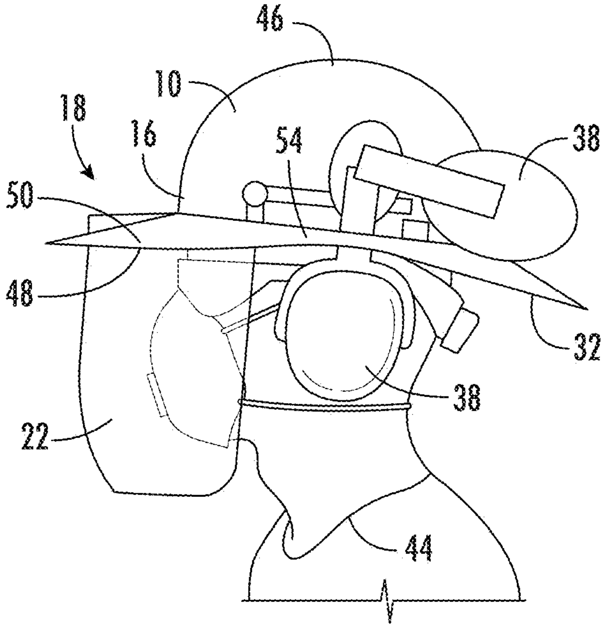


FIG. 6

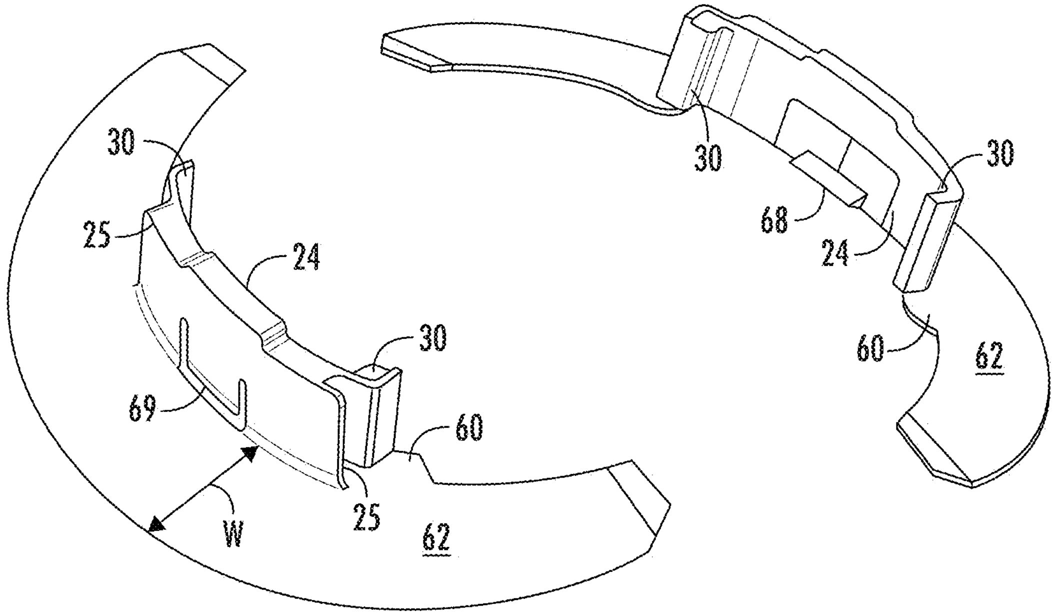


FIG. 7

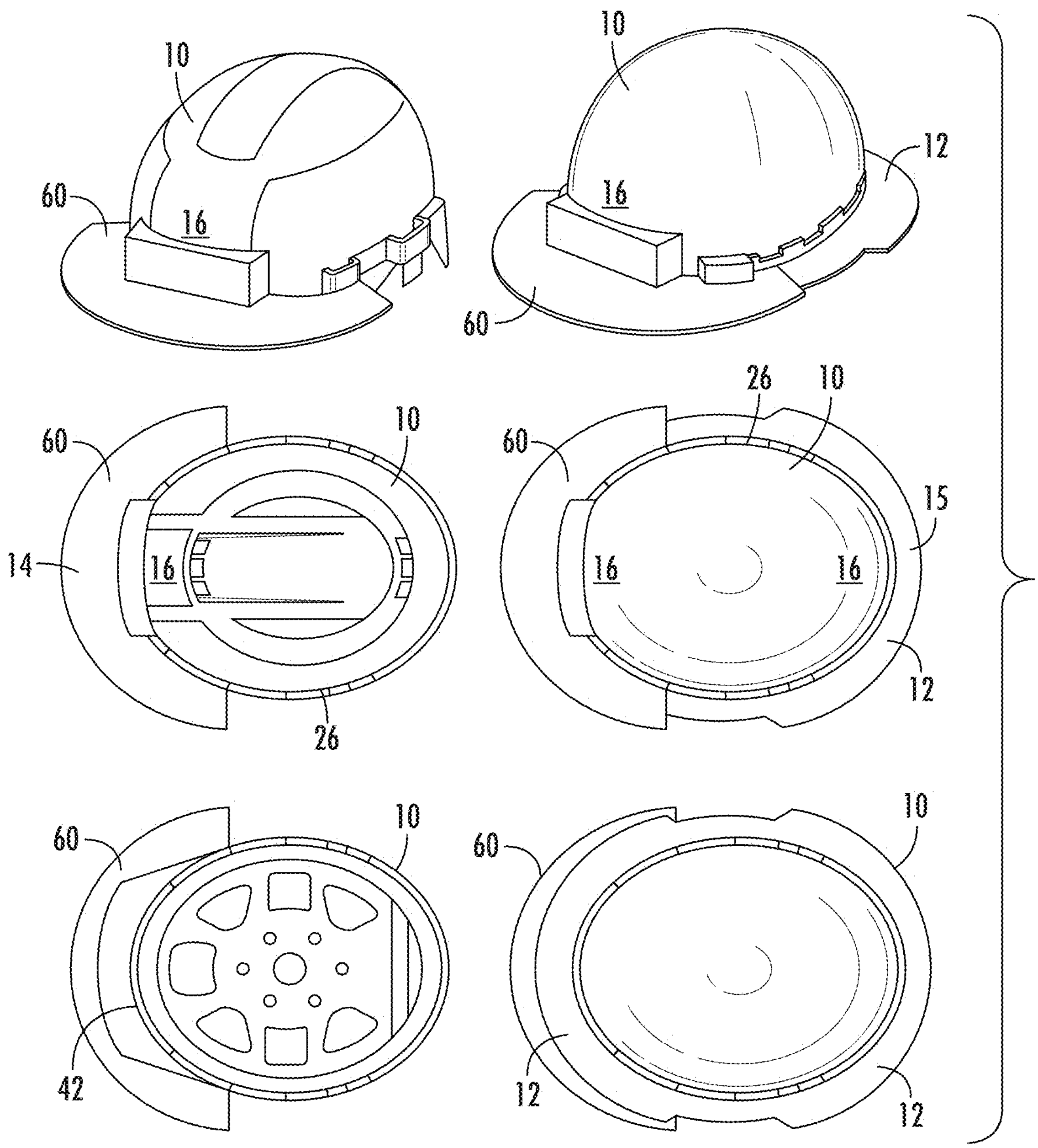

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FIG. 8

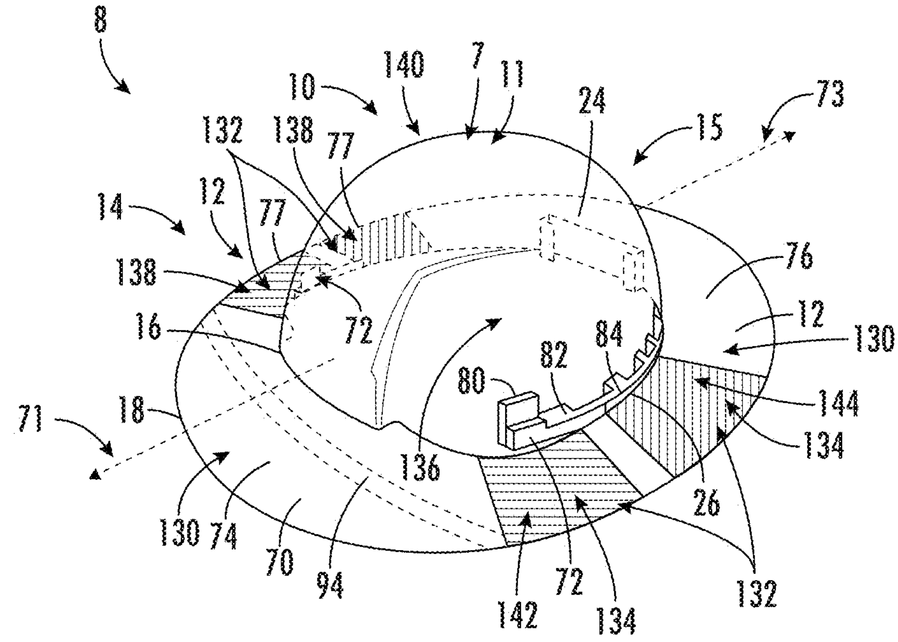
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FIG. 9

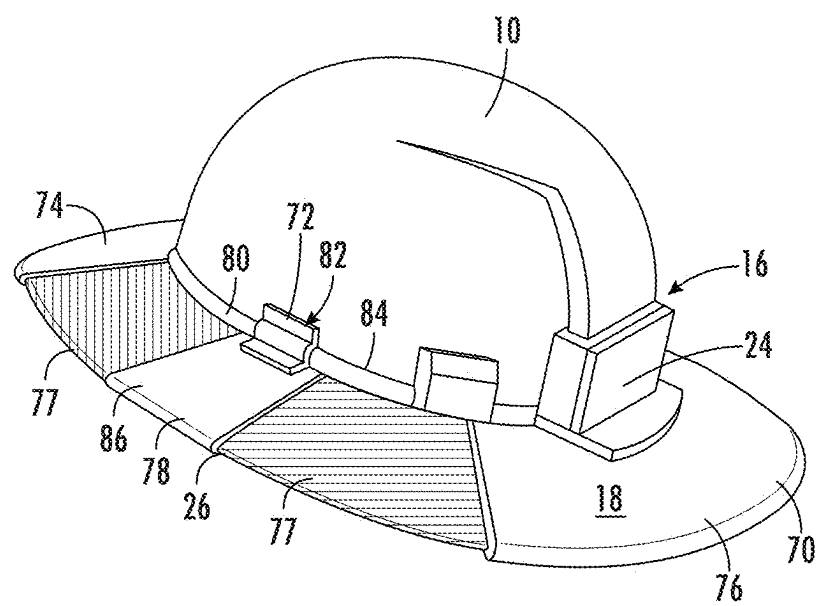
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FIG. 10

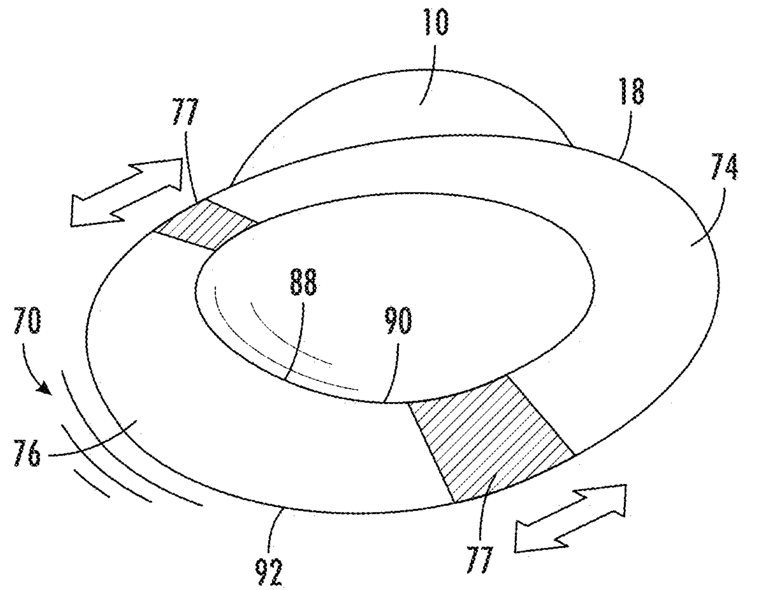

FIG. 11A

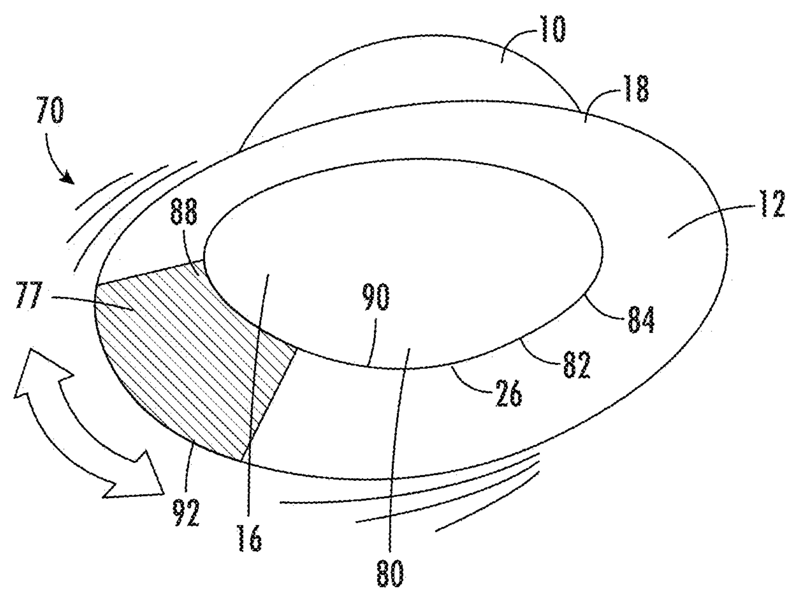

FIG. 11B

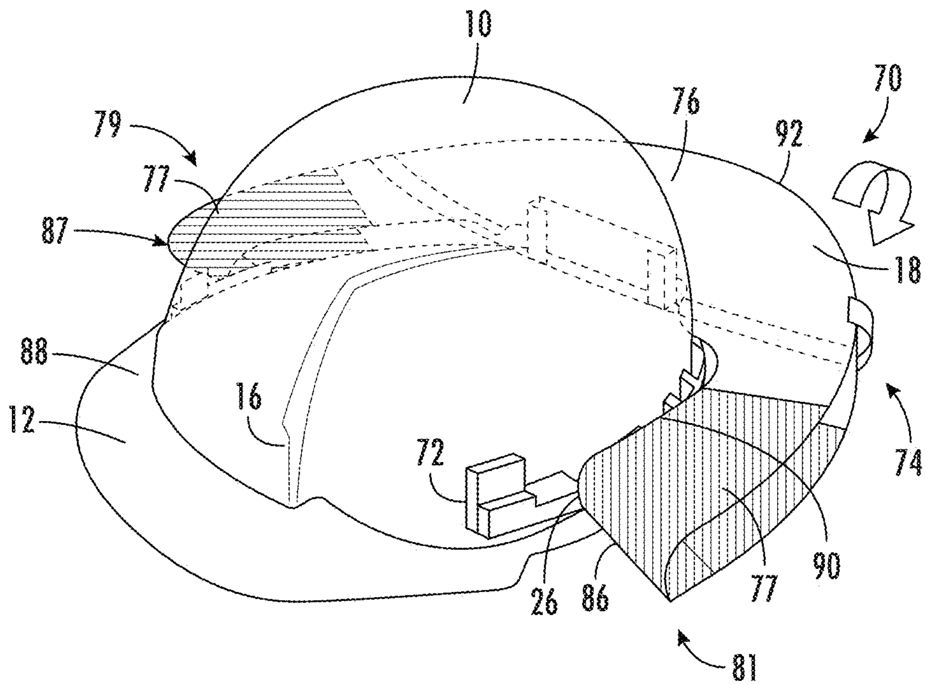
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FIG. 12

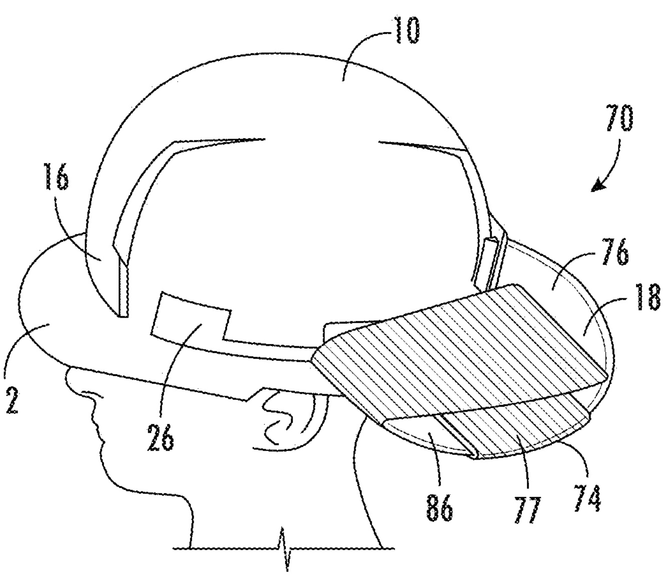
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FIG. 13

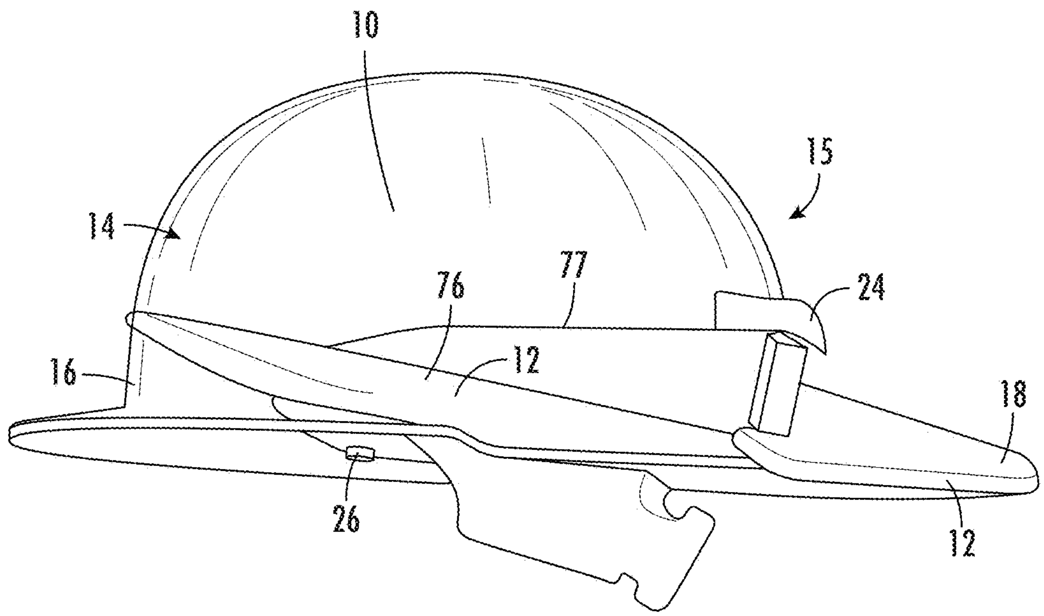
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FIG. 14

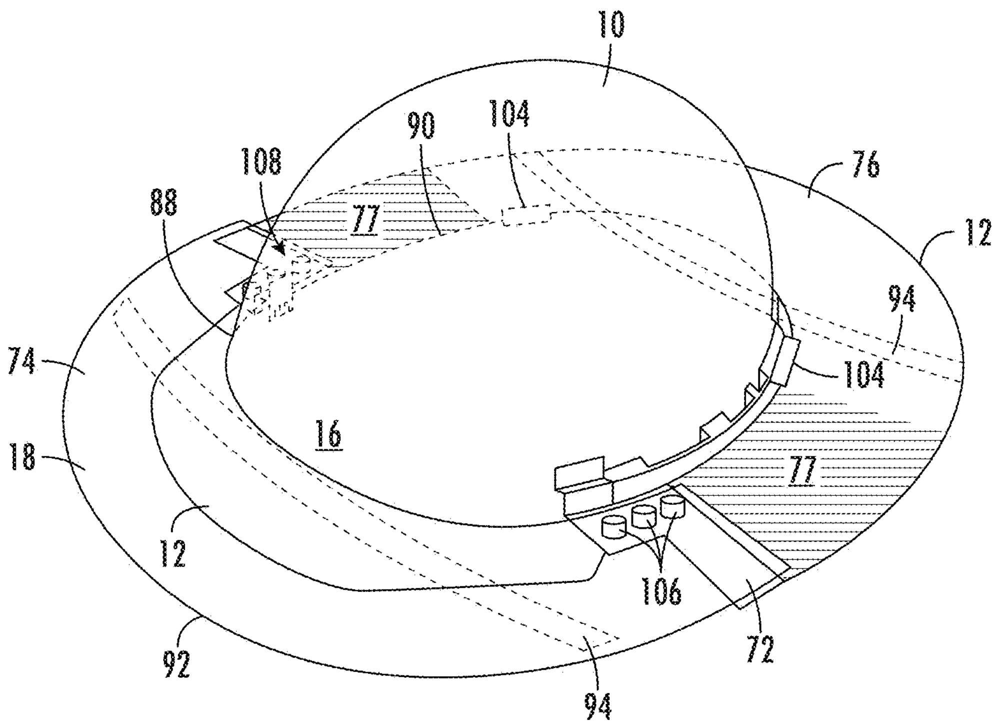
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FIG. 15

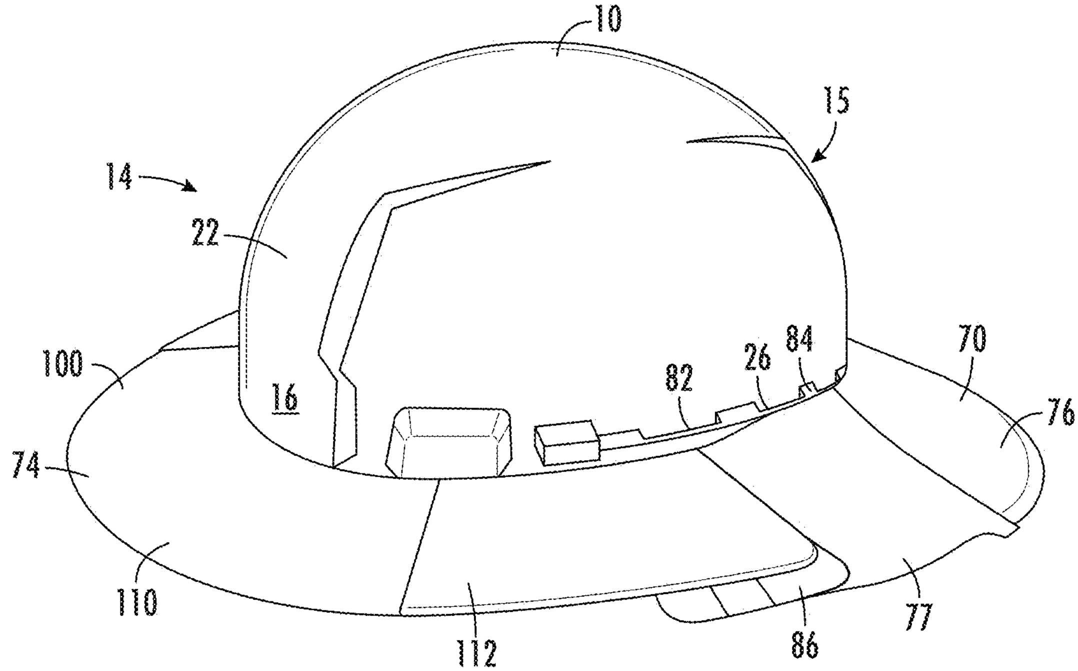
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FIG. 16

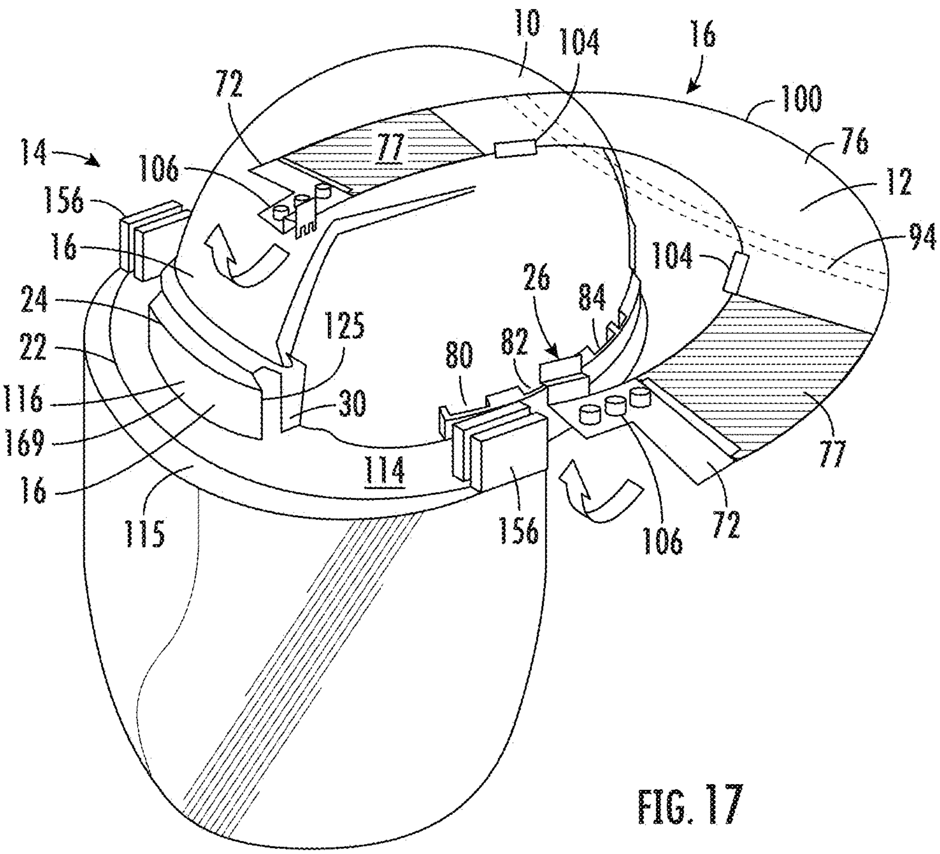
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FIG. 17

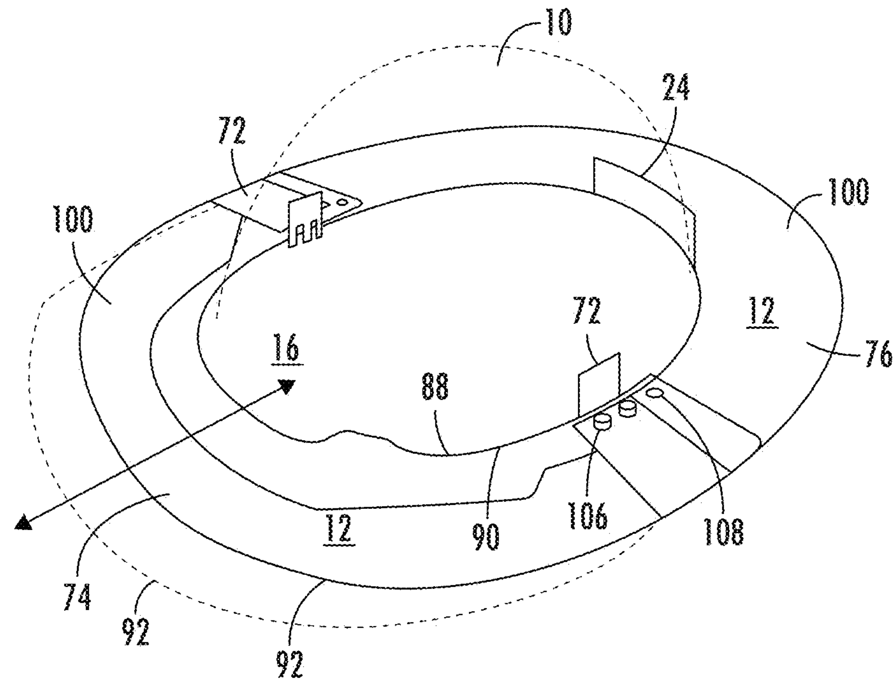
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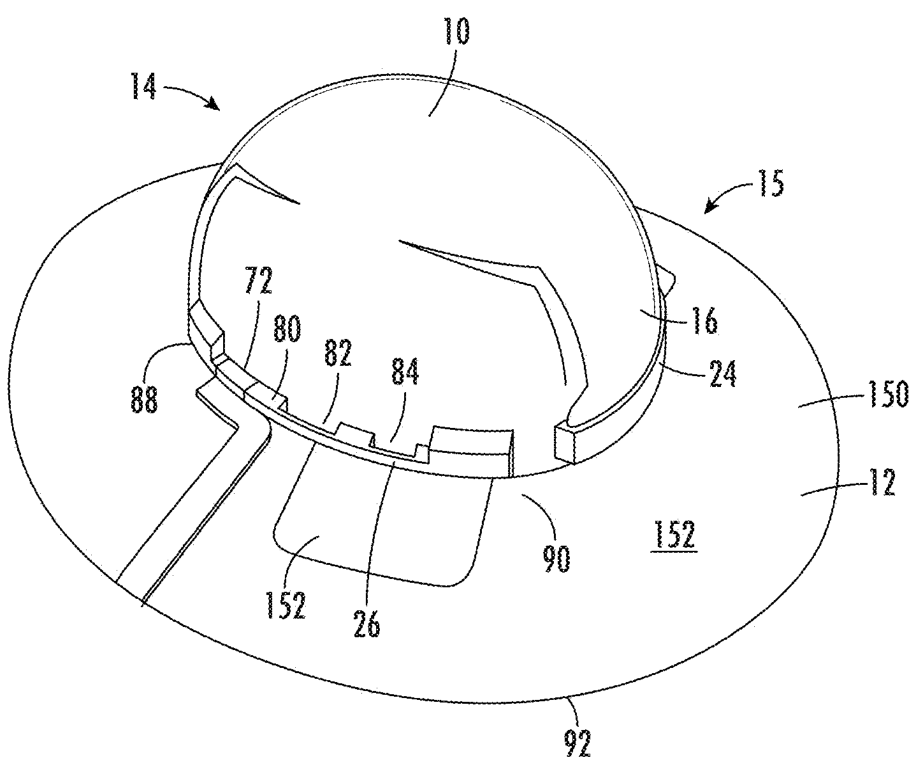
FIG. 18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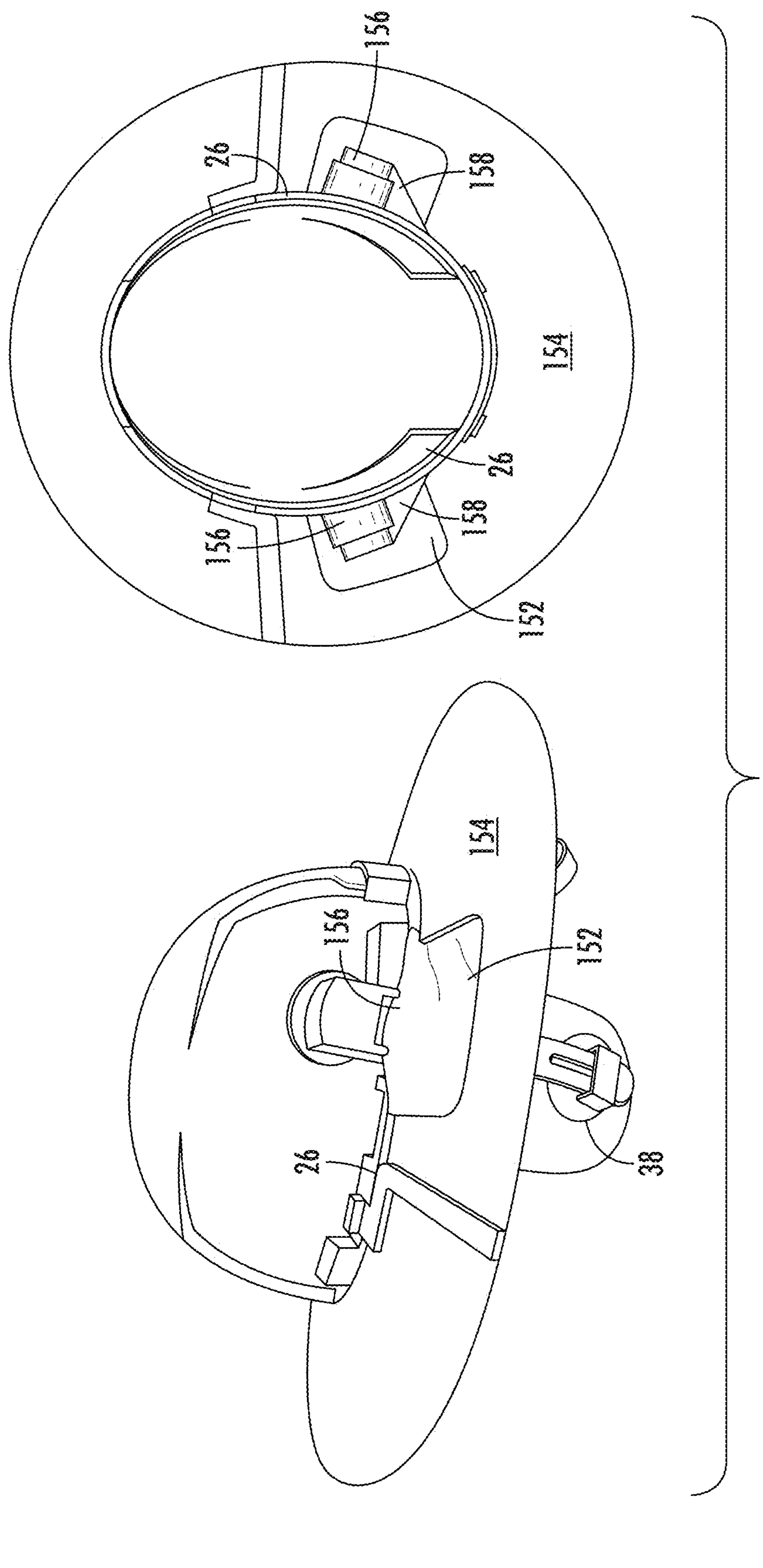

FIG. 20

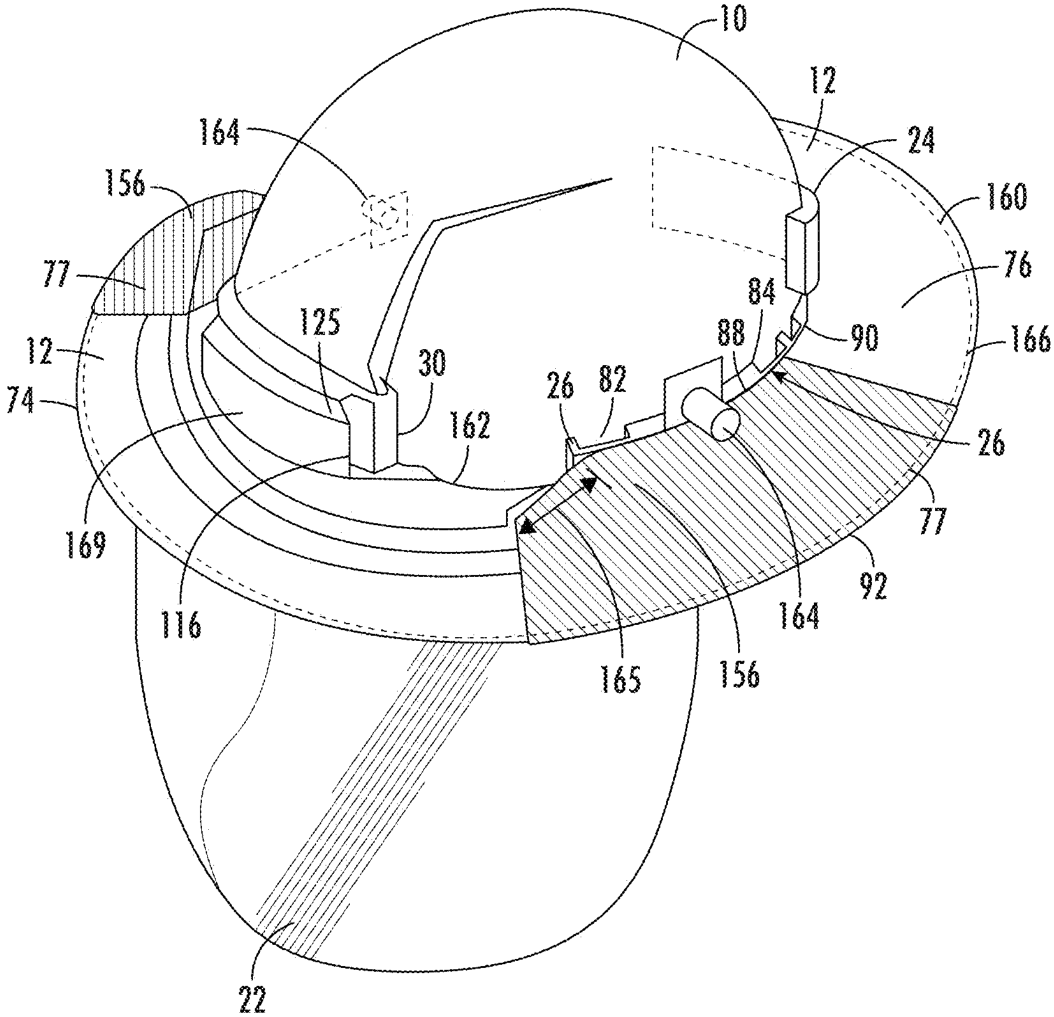
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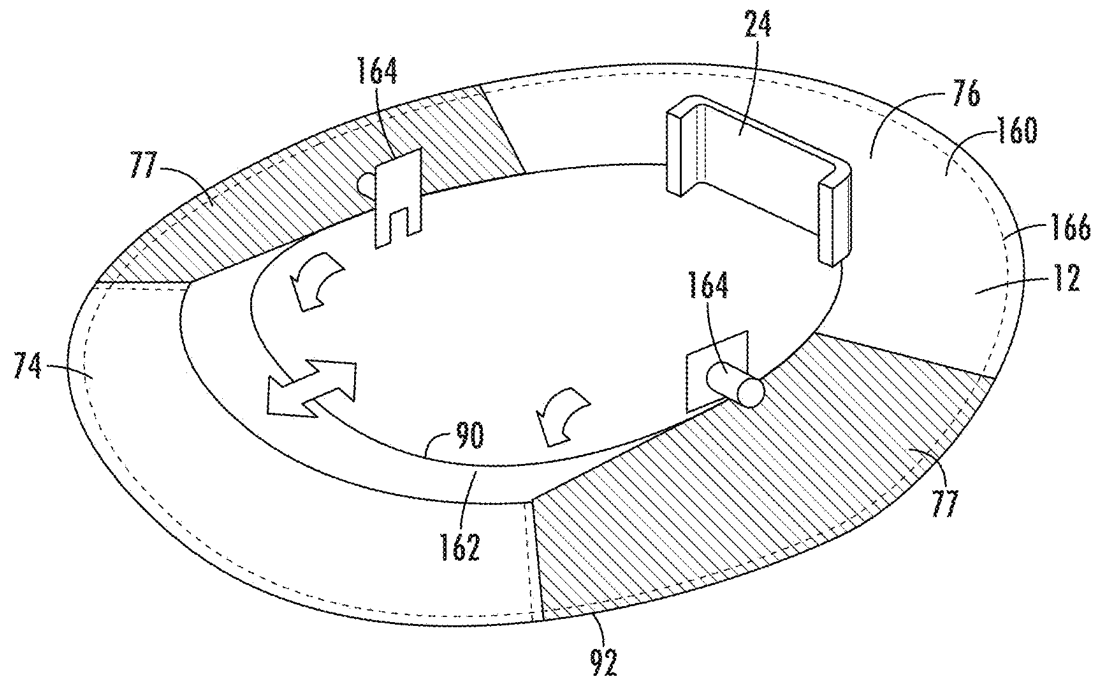
FIG. 21



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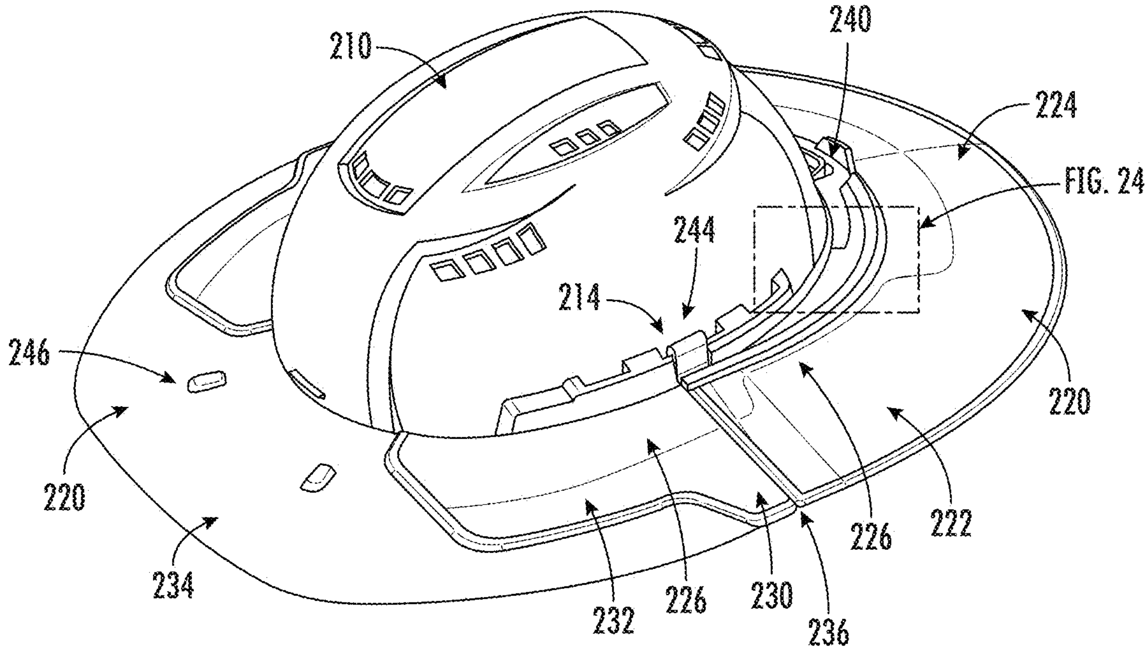
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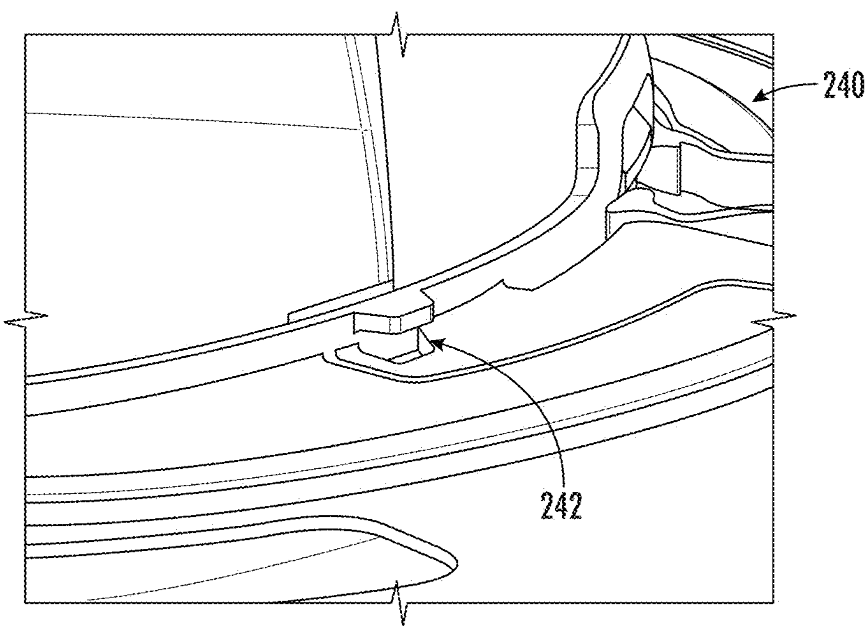
FIG. 23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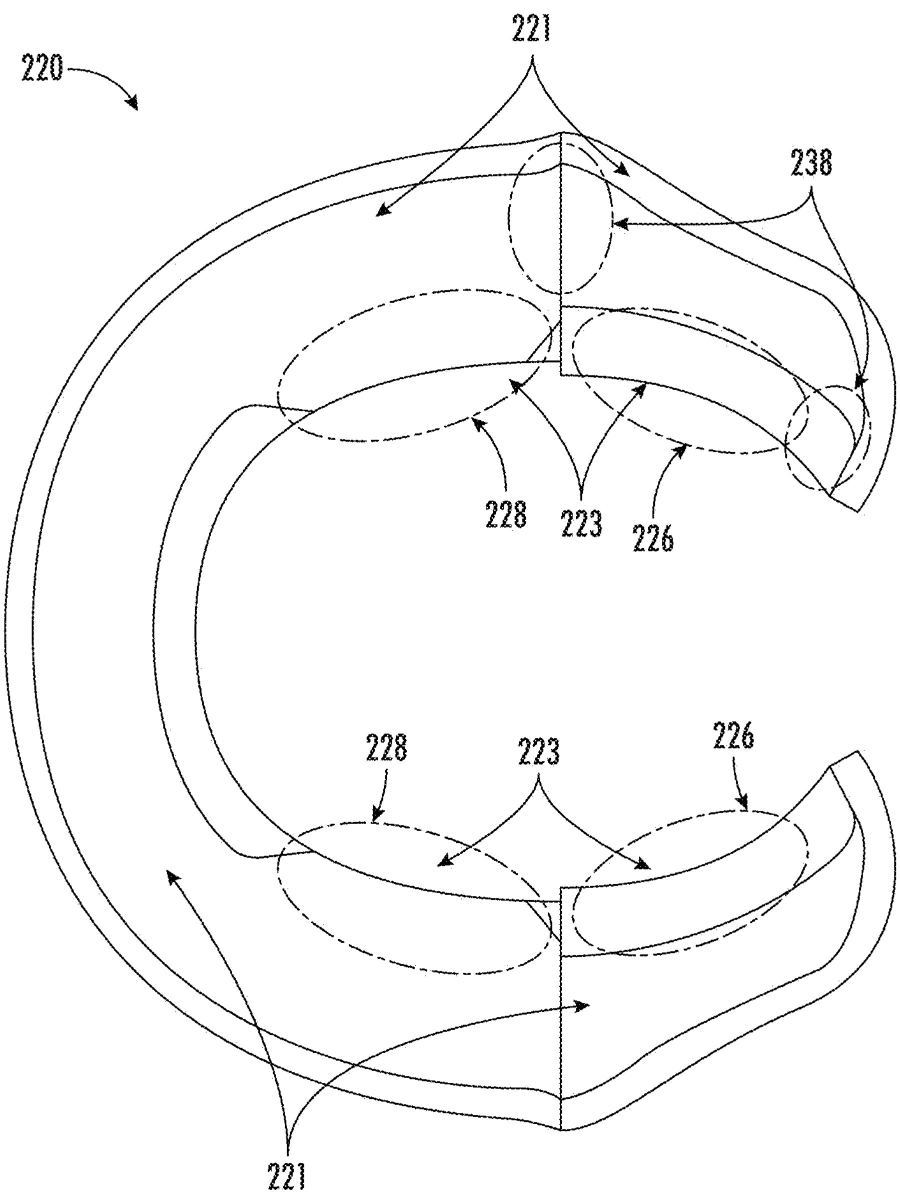

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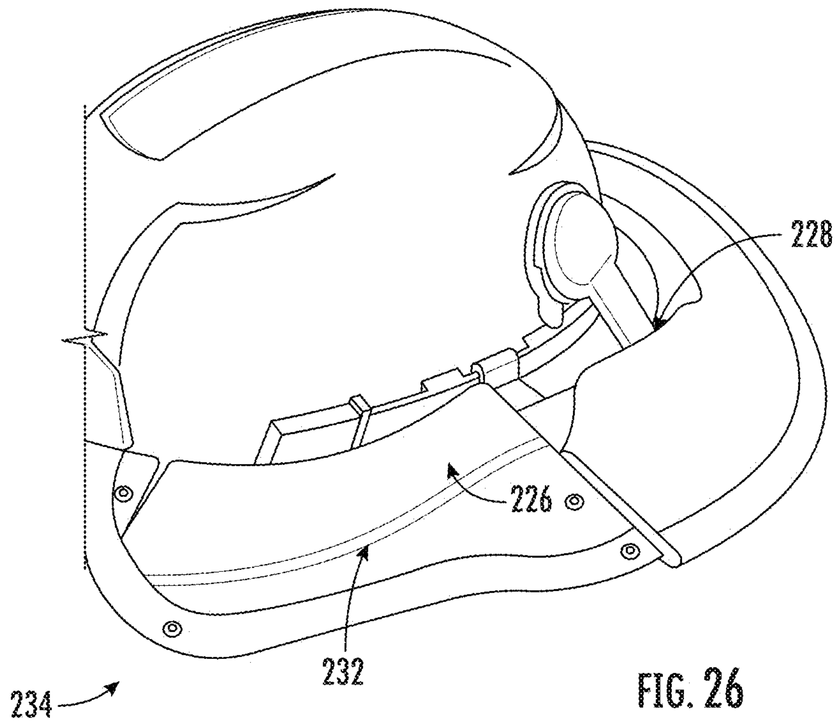

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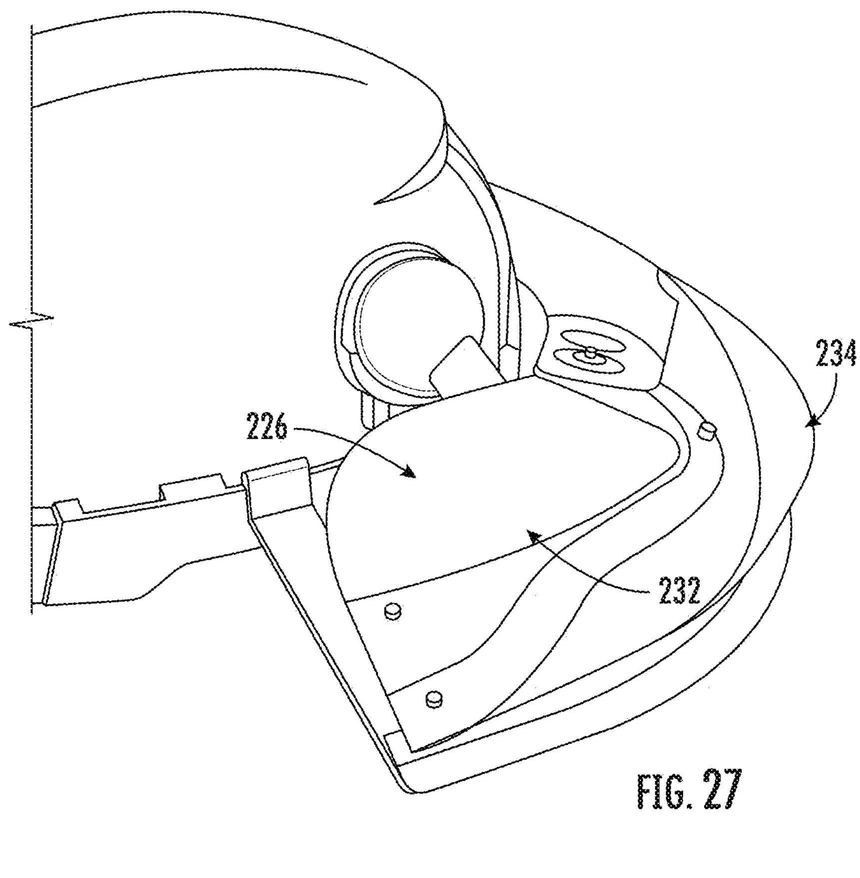

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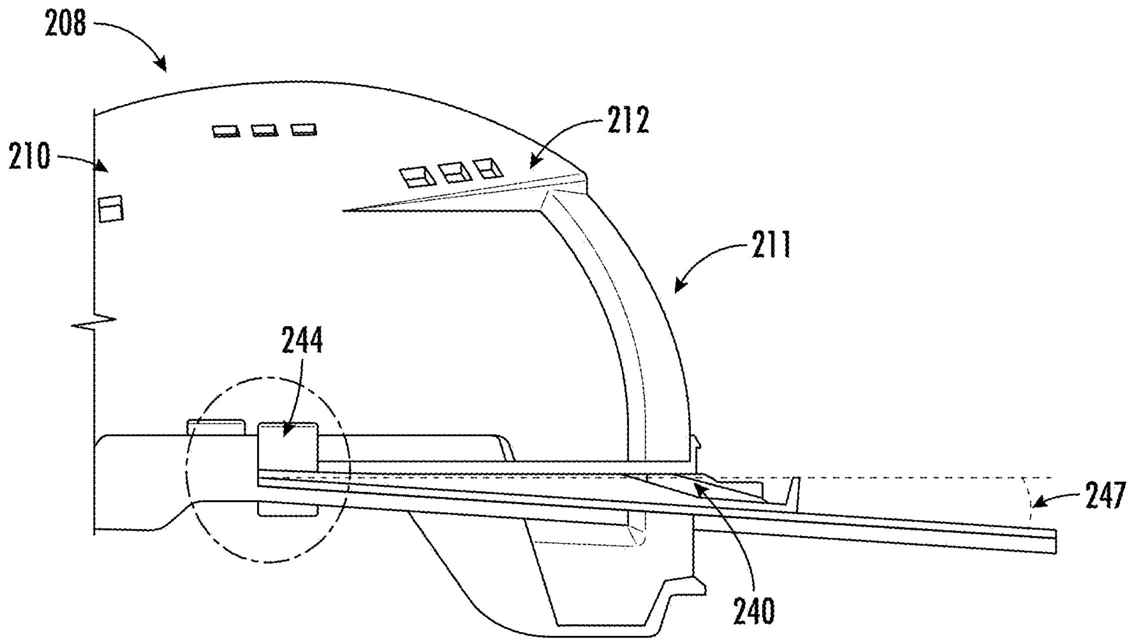

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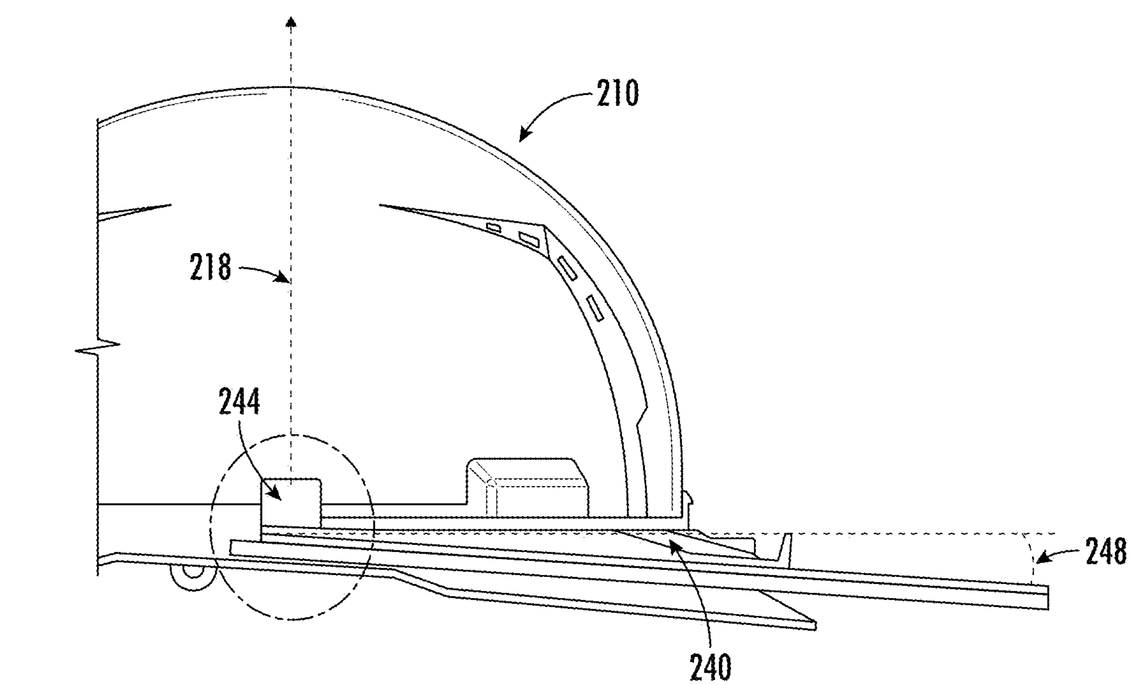

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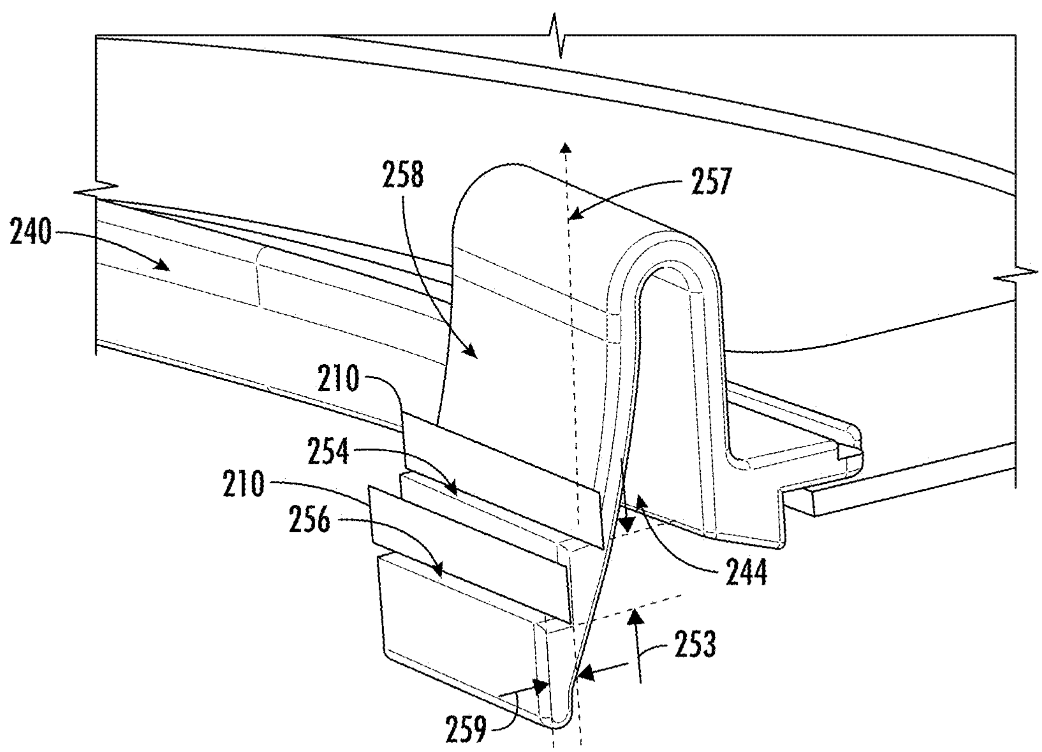
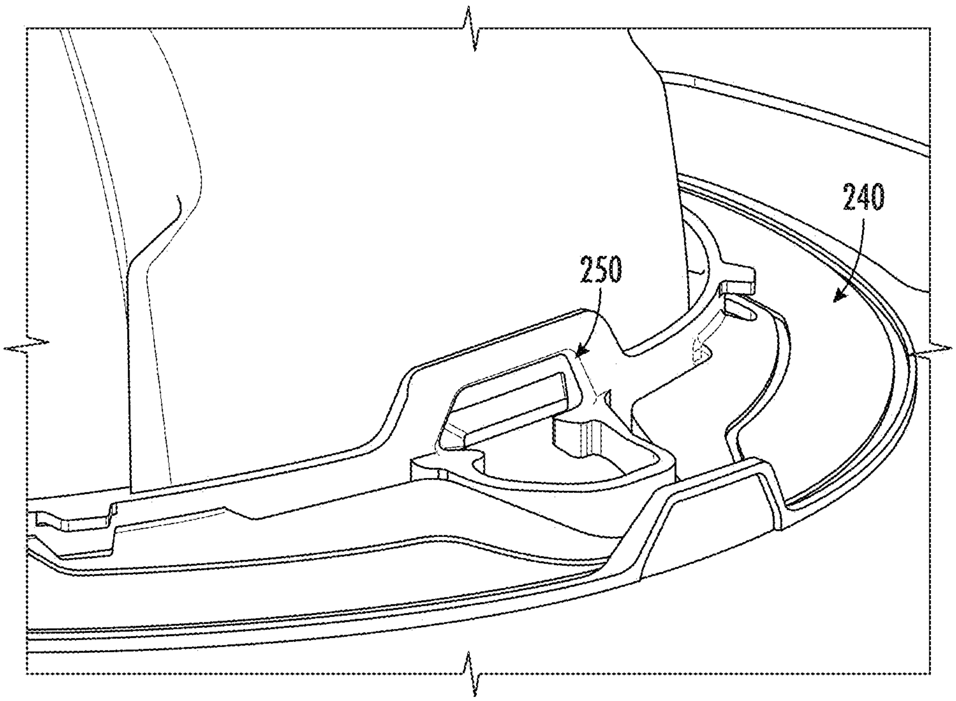


FIG. 30



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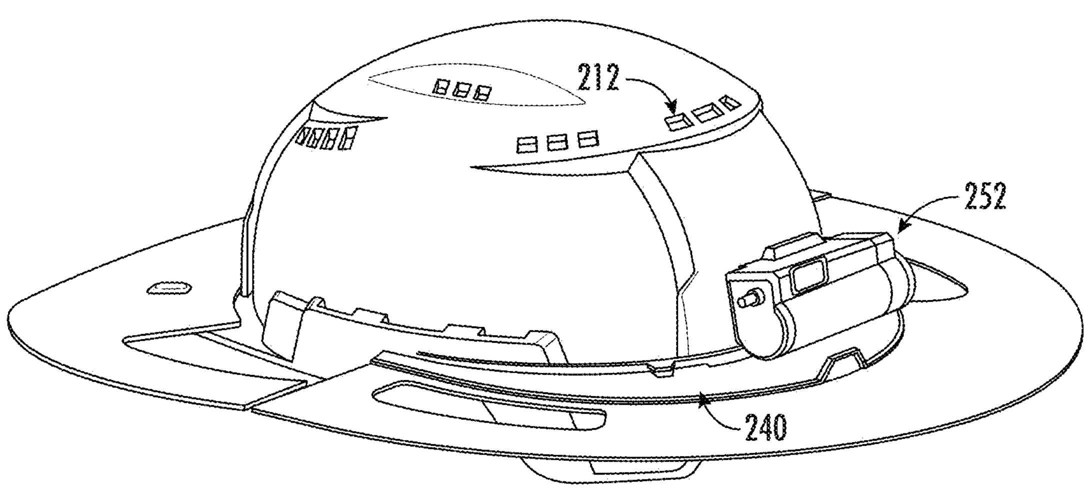


FIG. 32

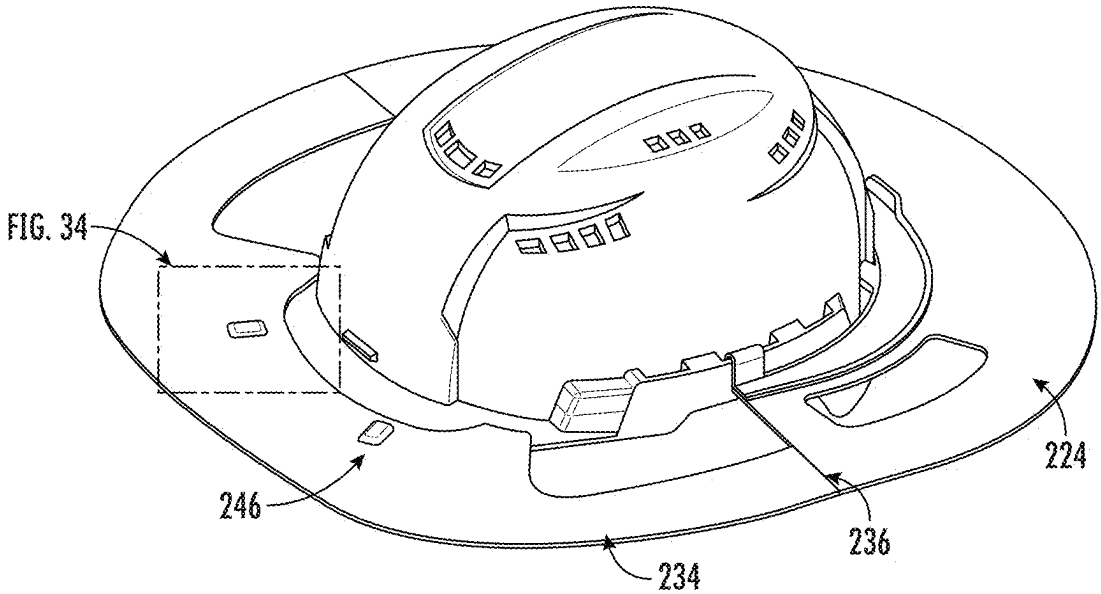


FIG. 33

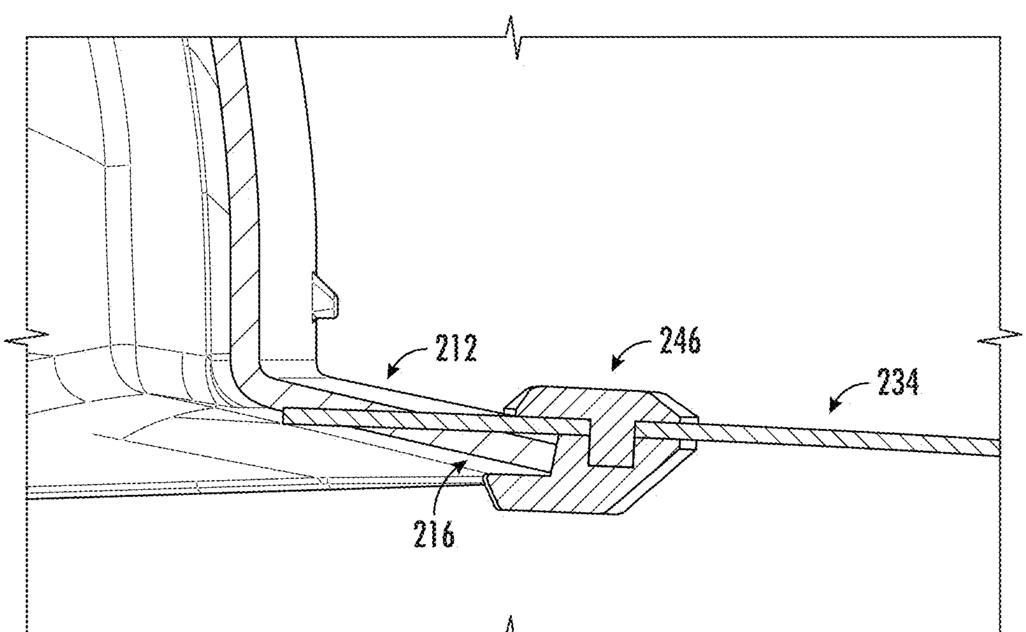
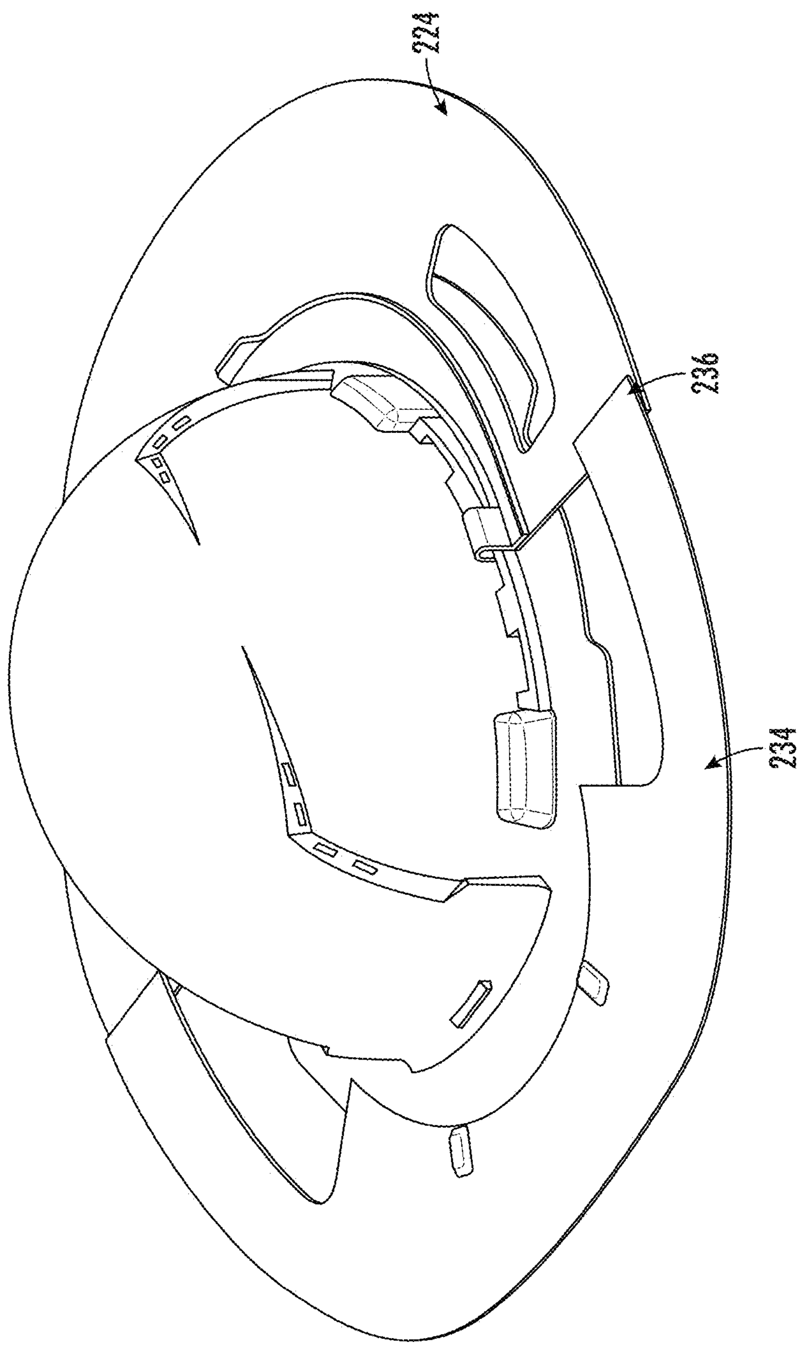


FIG. 34



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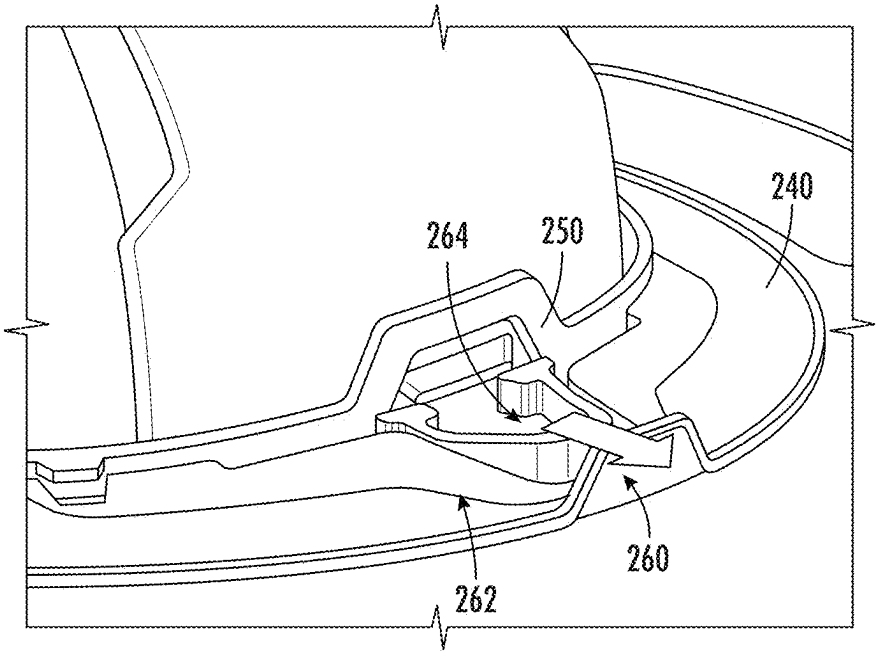

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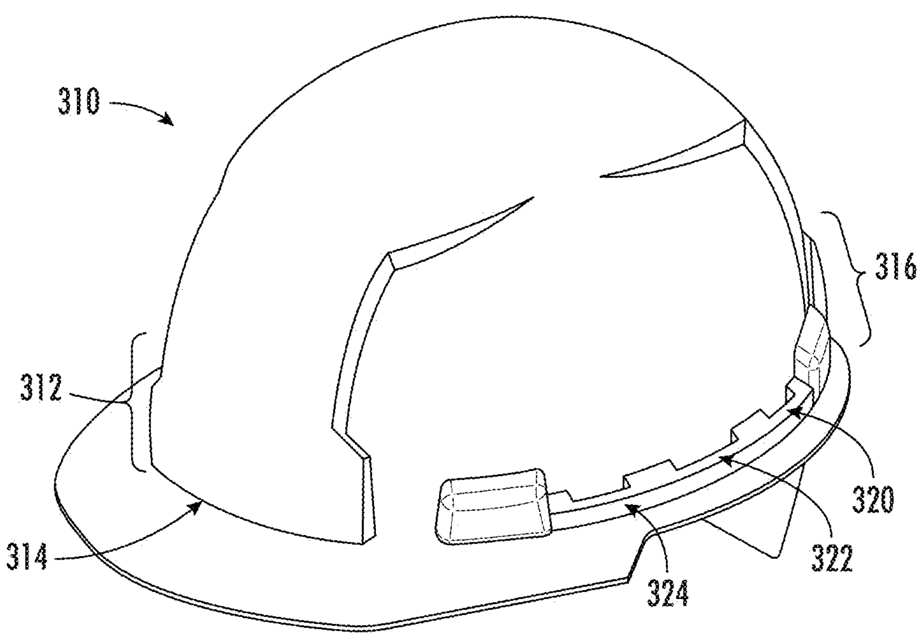

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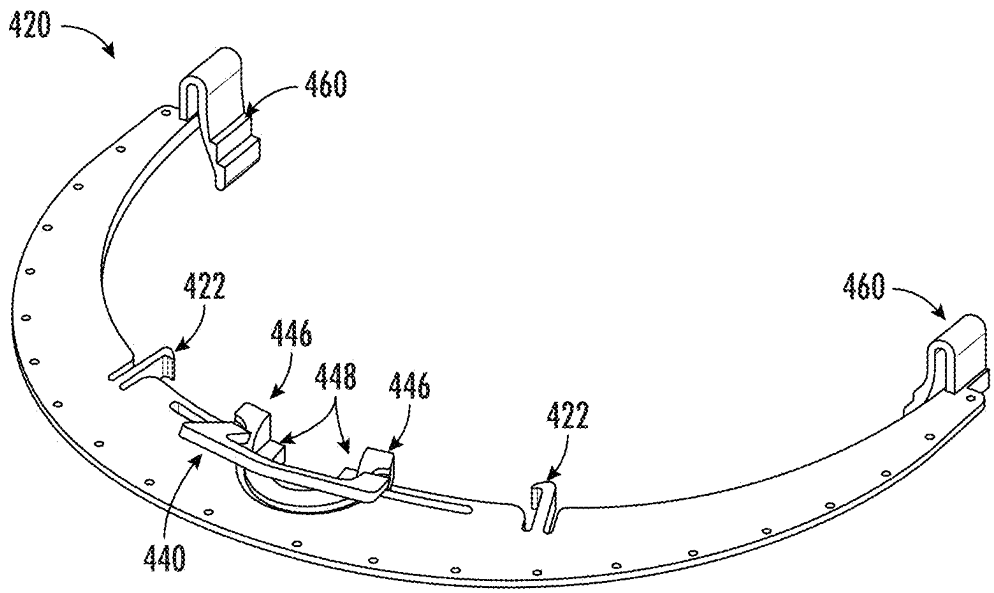

FIG. 38

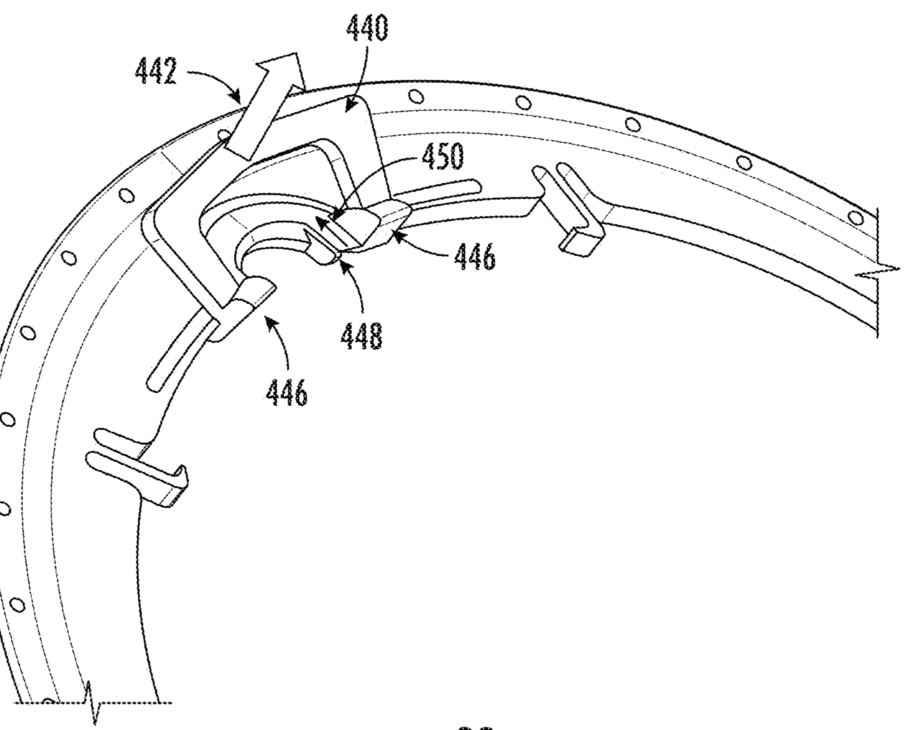

FIG. 39

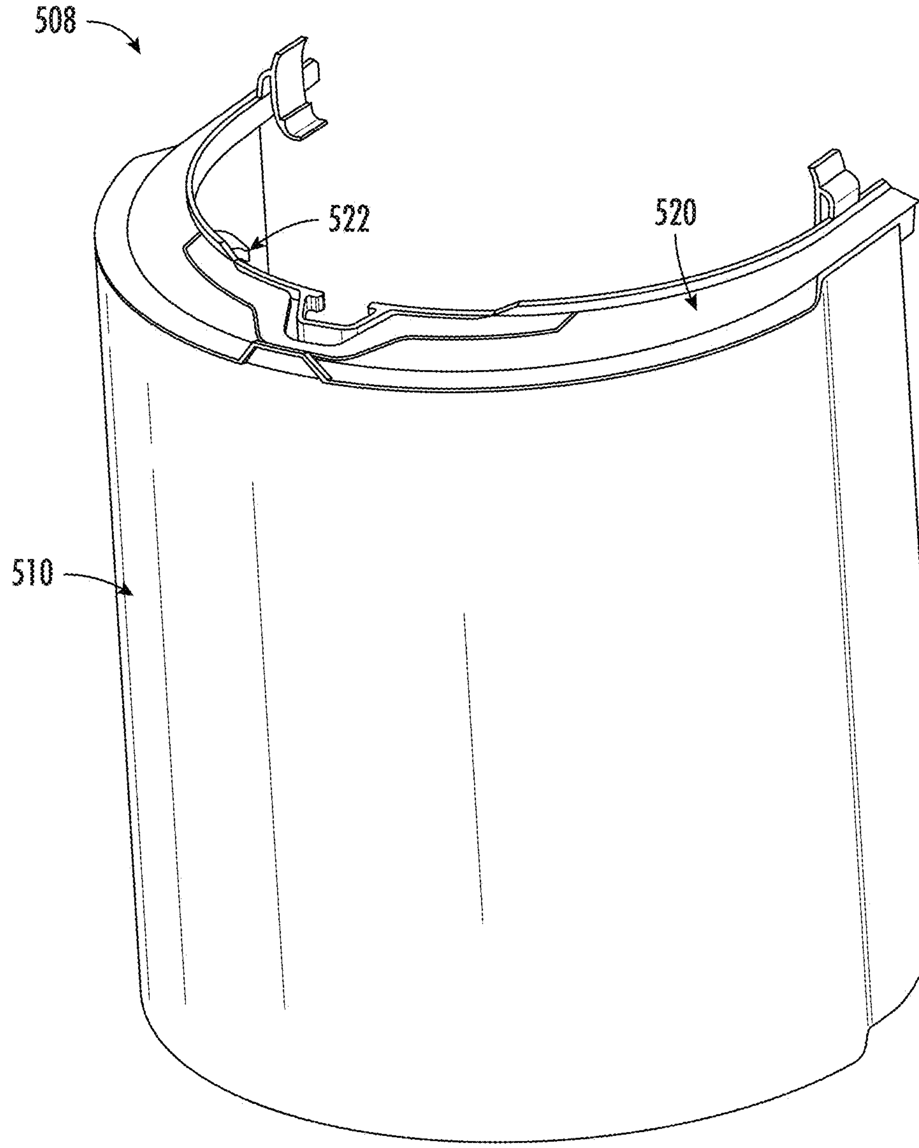

FIG. 40

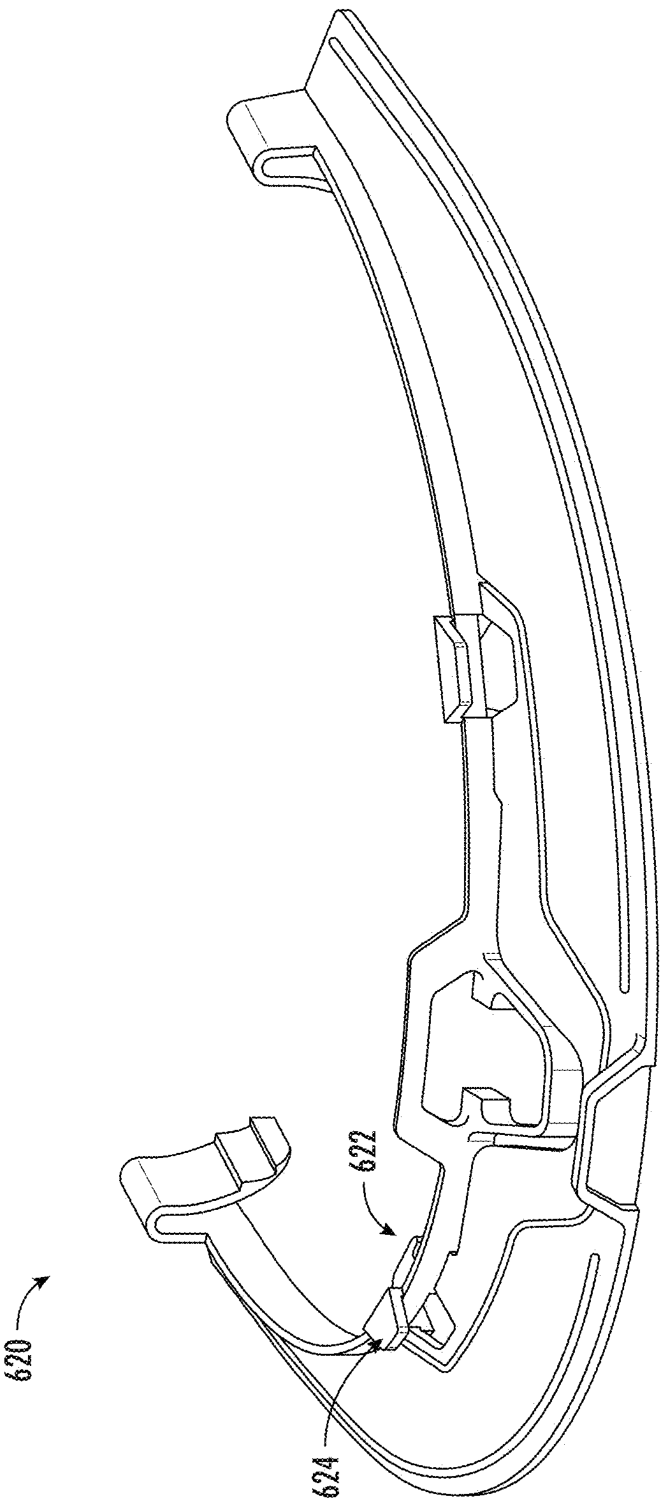

FIG. 41

HARD HAT ATTACHMENT SYSTEM AND SUN VISOR

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS

[0001] The present application is a continuation of U.S. application Ser. No. 17/461,163, filed Aug. 30, 2021, which is a continuation of International Application No. PCT/US2021/045405, filed Aug. 10, 2021, which claims the benefit of and priority to U.S. Provisional Application No. 63/167,458, filed on Mar. 29, 2021, U.S. Provisional Application No. 63/162,736, filed on Mar. 18, 2021, U.S. Provisional Application No. 63/087,578, filed on Oct. 5, 2020, and U.S. Provisional Application No. 63/066,561 filed on Aug. 17, 2020, each of which is incorporated herein by reference in their entirety.

BACKGROUND OF THE INVENTION

[0002] The present invention relates generally to the field of hard hats and helmets. The present invention relates specifically to a sun-visor and/or other safety equipment accessories attached to a hard hat. Hard hats are often used in loud and brightly illuminated areas, for example, a construction site on a sunny day. Work environments at various job sites may enable the use of added safety equipment (e.g., earmuffs and/or a face protector) to provide added protection to the user. A sun-visor may protect the user's face and/or neck from sunburn and/or falling overhead debris. A lamp or flashlight may assist with vision in poorly lit or low visibility environments or during a change from an illuminated to a dim work environments. Additional ear protection or earmuffs help protect the user's hearing in noisy environments.

SUMMARY OF THE INVENTION

[0003] One embodiment of the invention relates to a hard hat system including a hard hat and a visor. The hard hat includes a shell formed from a rigid material and an accessory mounting location located along an outer surface of the shell. The visor is reversibly coupled to the hard hat. The visor includes a first portion and a second portion. The first portion extends circumferentially around at least a portion of the shell and extends radially outward from the shell forming a sun-blocking flange. The second portion is coupled to the first portion and is positioned adjacent to the accessory mounting location. The second portion is deformable relative to the first portion such that the second portion deforms to accommodate and cover a portion of an accessory coupled to the accessory mounting location allowing the accessory to extend from the accessory mounting location to below the second portion.

[0004] Another embodiment of the invention relates to a hard hat system including a hard hat and a visor. The hard hat includes a shell formed from a rigid material. The visor is reversibly coupled to the hard hat. The visor includes a front portion and a rear portion. The front portion is moveable between a covering configuration in which the front portion extends in a forward direction away from a front of the hard hat, and a stowed configuration in which the front portion extends in a rearward direction away from a rear of the hard hat.

[0005] Another embodiment of the invention relates to a hard hat system including a hard hat comprising a shell formed from a rigid material, a clip coupled to the hard hat, and a visor. The clip includes an engaging portion including an upper surface and a lower surface. Each of the upper surface and the lower surface face in an upward direction. The clip is configured to couple to the hard hat in a first position or in a second position. The upper surface engages with the hard hat when the clip is in the first position and the lower surface engages with the hard hat when the clip is in the second position. The upper surface is spaced a distance above the lower surface. The visor is coupled to the clips such that the visor is reversibly supported to the hard hat via the clip and extends circumferentially around a portion of the hard hat.

[0006] One embodiment of the invention relates to a hard hat attachment system. The hard hat includes a mounting location comprising opposite dovetailed ridges projecting outward on either side of the mounting location. Each ridge includes a transition to couple slots of a bracket and support an accessory (e.g., lamp, face shield, and/or visor). Each ridge extends outwardly from the mounting location to create projections that support the mounted accessory. A sun-visor, e.g., an extended brim or visor, is coupled to the hard hat at the mounting location or the extended ridge to shade a wear's eyes, face and/or neck, to protect the user from sun/UV rays and/or falling debris, etc.

[0007] Another embodiment of the invention relates to a visor configured for selective attachment to a hard hat. The visor includes an inner and outer diameter. The visor includes a front brim and a rear brim. The front brim is positioned to extend outward from a front surface of the hard hat, and the rear brim is positioned to extend outward from a rear surface of the hard hat. A stretch zone extends radially from the inner diameter to the outer diameter and is located between and interconnects the front and rear brims. A fastener is coupled to the stretch zone and is configured to couple to an accessory ridge on the hard hat to removably secure the visor to the hat. In some embodiments, the visor includes a bracket positioned to engage one or more mounting locations on the hard hat to secure the visor to the hard hat. In a specific embodiment, the bracket is positioned on the front or rear visor to engage a front or rear mounting ridge of the hard hat.

[0008] In various embodiments, the visor includes a single stretch zone allowing an outer diameter of the visor to expand to accommodate different sized hard hats. In this embodiment, the single stretch zone is located between opposing ends of two visor sections allowing the size/circumference/inner diameter of the visor to be adjusted to accommodate different sized hard hats.

[0009] In various embodiments, the front brim includes a first end facing a first end of the rear visor and a second end facing a second end of the rear visor. In this arrangement, a first stretch zone is coupled between the first ends of the front and rear visors, and a second stretch zone is coupled between the second ends of the front and rear visors. In this arrangement, by having a plurality of stretch zones, e.g., on opposite sides of the visor sections, a parallel stretch sun-visor is provided that stretches in a direction between the front brim and the rear brim allowing for a distance between front and rear visors be expanded between a minimum distance and a maximum distance. In this configuration, the flexibility of the material of the stretch zones also provides for folding locations located on each side of the visor between the front and rear visors. This allows the user to fold the front brim over onto the back brim, and vice versa, to customize the brim location of the visor.

[0010] Another embodiment of the invention relates to a modular visor. The modular visor has a first module or brim section or segment that partially extends around the hard hat's perimeter. The modular visor enables a user to customize combinations of brim segments and/or accessories. For example, the user couples the rear modular brim section to a face shield in the front of hard hat. The user customizes a first modular brim section made from a semi-transparent material in front and couples it to a second modular brim section made from an opaque material in back. The modular sun-visor is used in variation with other accessories and/or segments. Segmentation of the modular visor enables the user to combine different modules made from different materials or features.

[0011] Another embodiment of the invention relates to a hard hat visor including enclosed elastic areas within a brim of the visor. The enclosed clastic areas extends radially outwards from the inner diameter of the visor towards the outer diameter and are located on opposite lateral sides of the brim. Specifically, the enclosed elastic areas are located adjacent to the auxiliary ridges of the hard hat and extend outwards from the auxiliary ridge a radial distance that is less than the distance to outer diameter. As such, the elastic areas do not extend across the entire radial length of the visor between the inner and outer edges and form an elastic inner brim coupled to a rigid outer brim. The elastic area expands or deflects when accessories are attached to the auxiliary ridge. For example, support arms of earmuffs or face shields are coupled to ports in the auxiliary ridge by deflecting elastic areas, but otherwise without removing or adjusting the visor. In this way, the earmuffs are removed and rotated off the user's ear without removing or adjusting either the hard hat or the visor.

[0012] Another embodiment of the invention relates to a sun-visor with a front brim and a rear brim coupled with two opposite stretch zones. The sun-visor has a wire that extends along the outer diameter. The front brim includes a semi-transparent material and a pad along the inner diameter of the front brim. The pad creates a buffer that deflects for different sizes or styles of hard hats. The stretch zones couple to the auxiliary ridge at a pivot joint that biases the visor about an outer perimeter of the hard hat. The stretch zones and pad deflect to accept variations in sizes for accessories attached in the mounting locations or ports of the auxiliary ridge without removing or adjusting the visor. The pad fits over and/or around additional accessories attached at mounting locations of the hard hat (e.g., a face shield). The visor folds about a spring-loaded joint or biased pivot to adjust a frictional clamping force of an inner diameter of the visor against the outer perimeter of the hard hat. The biased pivot also includes a joint that couples the sun-visor to a mounting location or auxiliary ridge of the hard hat.

[0013] In various additional embodiments, the sun-visor is configured for any style or size of helmet or hard hat (e.g., a traditional or climbing style) that includes the mounting locations and/or auxiliary ridge. In this way a universal mounting system that includes mounting locations and auxiliary ridges is compatible with a variety of accessories and configured to work with the visors described herein. The visor can include a bracket that couples to a front or rear mounting location. For example, a notch located on a face of the mounting location secures and removably locks the bracket against the mounting location. For example, the notch can fit within an inner recess of bracket to lock bracket against the notch at the mounting location. In another embodiment, the visor slides under the notch along the outer perimeter of the hard hat and bracket is coupled over the visor to compress visor and securely lock the visor under the notch.

[0014] Alternative exemplary embodiments relate to other features and combinations of features as may be generally recited in the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] This application will become more fully understood from the following detailed description, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements in which:

[0016] FIG. **1** is a hard hat with a forward accessory mounting location worn with the brim facing forward, according to an exemplary embodiment.

[0017] FIG. **2** is a hard hat with a rearward accessory mounting location worn with the brim facing backward, according to an exemplary embodiment.

[0018] FIG. **3** is a detailed view of a disconnected lamp accessory connected to a rigid bracket at the forward mounting location, according to an exemplary embodiment.

[0019] FIG. **4** is a comparison of a bill style hard hat and a brim hard hat supporting earmuffs, according to an exemplary embodiment.

[0020] FIG. **5** is a cross-sectional comparison of the bill style hard hat and the brim hard hat of FIG. **3**, according to an exemplary embodiment.

[0021] FIG. **6** shows the mounting locations of hard hat to mount hardware (e.g., earmuffs) over the neck foundation, according to an exemplary embodiment.

[0022] FIG. **7** is two perspective views of a modular visor, according to an exemplary embodiment.

[0023] FIG. **8** shows how the visor of FIG. **7** couples to both the bill style and brim hard hats, according to an exemplary embodiment.

[0024] FIG. **9** is a top perspective view of a visor with a bracket over an outer perimeter of a hard hat, according to an exemplary embodiment.

[0025] FIG. **10** is a perspective side view of the sun-visor of FIG. **9**, according to an exemplary embodiment.

[0026] FIG. **11**A is a perspective view of a visor with a parallel stretch visor with two stretch zones, according to an exemplary embodiment.

[0027] FIG. **11**B is a perspective view of a visor with one stretch zone, according to an exemplary embodiment.

[0028] FIG. **12** is a top perspective view of the parallel stretch visor of FIG. **9** in a folded configuration, according to an exemplary embodiment.

[0029] FIG. **13** is a top side view of the visor of FIG. **9** in a folded configuration, according to an exemplary embodiment.

[0030] FIG. **14** is a side view of the visor of FIG. **9** in a folded configuration, according to an exemplary embodiment.

[0031] FIG. **15** is a perspective view of another embodiment of a modular sun-visor, according to an exemplary embodiment.

[0032] FIG. **16** shows the modular sun-visor and the folded sun-visor in the front and rear of the hard hat, respectively, according to an exemplary embodiment.

[0033] FIG. **17** is a top perspective view of a modular visor coupled to a face shield accessory, according to an exemplary embodiment.

[0034] FIG. **18** is a perspective view of a modular visor having a front semi-transparent material segment coupled to a rear opaque material segment; the modular visors are coupled to each other and to the hard hat with a fastened joint, according to an exemplary embodiment.

[0035] FIG. **19** is a perspective view of an enclosed stretch elastic area within the brim of a visor and located adjacent to the auxiliary ridges of hard hat, according to an exemplary embodiment.

[0036] FIG. **20** is a perspective views of the enclosed elastic area of the visor embodiment of FIG. **19**, according to an exemplary embodiment.

[0037] FIG. **21** is another embodiment of a multi material visor that includes a pad, a wire, and/or a bracket to adjustably couple the visor to different sizes and shapes of hard hats with various attached accessories (e.g., a face shield), according to an exemplary embodiment.

[0038] FIG. **22** is a perspective view of the visor of FIG. **21**, according to an exemplary embodiment.

[0039] FIG. **23** is a perspective view of a visor, according to an exemplary embodiment.

[0040] FIG. **24** is a detailed perspective view of the visor of FIG. **23**, according to an exemplary embodiment.

[0041] FIG. **25** is a top view of a portion of the visor of FIG. **23**, according to an exemplary embodiment.

[0042] FIG. **26** is a perspective view of the visor of FIG. **23**, according to an exemplary embodiment.

[0043] FIG. **27** is a perspective view of the visor of FIG. **23**, according to an exemplary embodiment.

[0044] FIG. **28** is a side view of the visor of FIG. **23** coupled to a helmet, according to an exemplary embodiment.

[0045] FIG. **29** is a side view of the visor of FIG. **23** coupled to a hard hat, according to an exemplary embodiment.

[0046] FIG. **30** is a detailed perspective view of a portion of the visor of FIG. **23**, according to an exemplary embodiment.

[0047] FIG. **31** is a detailed perspective view of a portion of the visor of FIG. **23**, according to an exemplary embodiment.

[0048] FIG. **32** is a perspective view of a portion of the visor of FIG. **23**, according to an exemplary embodiment.

[0049] FIG. **33** is a perspective view of the visor of FIG. **23**, according to an exemplary embodiment.

[0050] FIG. **34** is a cross-section view of a portion of the visor of FIG. **23**, according to an exemplary embodiment.

[0051] FIG. **35** is a perspective view of the visor of FIG. **23**, according to an exemplary embodiment.

[0052] FIG. **36** is a perspective view of the hard hat and visor of FIG. **31**, according to an exemplary embodiment.

[0053] FIG. **37** is a perspective view of a hard hat, according to an exemplary embodiment.

[0054] FIG. **38** is a perspective view of a frame, according to an exemplary embodiment.

[0055] FIG. **39** is a perspective view of the frame of FIG. **38**, according to an exemplary embodiment.

[0056] FIG. **40** is a perspective view of a hard hat system including a face shield and a frame, according to an exemplary embodiment.

[0057] FIG. **41** is a perspective view of a frame, according to an exemplary embodiment.

DETAILED DESCRIPTION

[0058] Referring generally to the figures, a hard hat accessory system is shown and described. Hard hats include both traditional brimmed style hard hats and climbing style helmets, e.g., with only a small bill in front. Unless specifically indicated, Applicant has used the term "hard hat" to include traditional and climbing style hard hats, as well as any hard hat or protective helmet design, including hard hats having a universal mounting system with mounting locations and auxiliary ridges. Different sizes and styles of hard hats present challenges to the uniform mounting system's size and dimensions. The universal mounting system includes both front and rear mounting locations and opposite side auxiliary ridges. The mounting system provides uniform sized attachments and locations to mount various accessories on various sizes and styles of hard hats. Applicant has found that a visor accessory mounted to the uniform mounting system can also provide the operator multiple access sites at the mounting ridges and side ports to mount additional personal/safety equipment. This enables a user to customize the hard hat for the job site and personal preferences, which enhances user compliance and the proper use of adequate safety equipment based on the job specifications. As used herein, accessories generally include lamps (or headlights), face shields, and/or earmuffs, but can also include a reflector, a magnetic tool carrier, or tool/fastener carrier, hand/power tools, and/or an eyeglass/safety glass holder.

[0059] A visor extends the brim of hard hat to shade the user's eyes, to protect the user's eyes from direct sunlight at a construction site, to block/shield the eyes/front of the face from light debris and dust, etc. Attachment of various additional accessories to the mounting system of hard hat can sometimes interfere with other previously mounted equipment. For example, an operator may frequently wear and take off earmuffs at the job site, which would require the operator to adjust both the visor and the earmuffs on/off the hard hat. Applicant has found that support on the mounting system for the rotation of earmuffs when not in use, e.g., off and behind the user's ears, enables the user to switch frequently wear and storage of the earmuffs. The ability to switch between a use and stored position enhances user comfort, particularly when the switch minimizes the adjustments to other attached accessories, such as a visor. Thus, in various embodiments, discussed below, Applicant has developed an innovative modular visor which includes relief portions, such as flexible expanding segments and/or portions that are less taut than neighboring portions, which allows the user to adjust earmuffs without needing to remove the visor. Applicant has also developed an innovative modular visor that permits the user to fold a front and/or side portion of the visor towards the rear, thus permitting the user to attach other accessories and/or attachments to a front and/or side of the hard hat without requiring the user to remove the visor.

[0060] FIG. **1** shows an operator wearing a hard hat **10**. FIG. **1** is a brim **12** styled or traditional hard hat **10** because a brim **12** surrounds and/or extends around a front **14** and rear **15** of hard hat **10**. As used herein, a brim **12** is an extension on or attached to hard hat **10** to protect from sun or debris. For example, a sun-visor serves to extend the brim

12 and create additional sun or debris protection. Hard hat **10** generally refers generically to both brim (e.g., traditional) and bill (e.g., climbing helmet) styled hard hats **10** unless expressly noted otherwise.

[0061] As illustrated, hard hat **10** is oriented in a forward-facing direction with a front mounting location **16** above a visor, ridge, bill, or brim **12** of the front **14** of hard hat **10**. In this configuration, front brim **12** is located at the front **14** of hard hat **10**, shields the operator's eyes, e.g., from the sun. In the illustrated position, a rear **15** of hard hat **10** is in the back of the user's head and provides a smaller brim **12** to shade to the user's neck.

[0062] FIG. **2** shows a reversed hard hat **10**, such that the front **14** is at the back of the user's head, and the rear **15** is above the eyes. A rear mounting location **16**, the same as or similar to the front **14** mounting location **16**, is reversed. In other words, the rear **15** mounting location **16** is located above the user's eyes and available for attachment of an accessory **18** (FIGS. **3** and **9**, e.g., a lamp **20** or face shield **22**) to hard hat **10**. In both FIGS. **1** and **2**, each attached accessory **18**, such as a headlamp and/or face visor, includes a mounting bracket **24** (FIG. **3**) that securely attaches to ridges **25** on front **14** or rear **15** mounting locations **16** of hard hat **10**.

[0063] FIG. **3** illustrates bracket **24** attachment of a headlamp or lamp **20** at mounting location **16** on hard hat **10**. In some embodiments, the attachment system includes mounting bracket **24**, lamp **20**, and a strap **28**. Bracket **24** includes clips or receiving slots **30** that interface with ridges **25** on front or rear mounting locations **16** of hard hat **10**. Bracket **24** provides structural support to lamp **20** and provides a rigid attachment location to secure lamp **20** on hard hat **10**. Strap **28** interconnects lamp **20** to bracket **24** and provides a mechanism to attach various lamps **20**, including after-market lamps with strap **28** at receiving support locations or mounting locations **16** of hard hat **10**.

[0064] A strap or band **32** wraps around the circumference of hard hat **10** without interfering with the mounting locations **16** or auxiliary ridges **26**. For example, band **32** passes through openings under auxiliary ridges **26**. Band **32** supports hand tools and/or other equipment suitable for storage along brim **12** or auxiliary ridges **26** of hard hat **10**. Similarly, various accessories **18** attach or couple to ports **34** in an auxiliary ridge **26** of hard hat **10**. In some embodiments, tools and/or other accessories **18** include an insert, fastener, or receiving slot **30** that couples with one or more ports **34**.

[0065] FIGS. **1** and **2** show different sized ports **34**. A smaller port **34** accommodates a smaller accessory **18** (e.g., tool or eyeglass carrier). Similarly, a larger port **34** is used for a larger accessory **18** (e.g., earmuffs **38**). In some embodiments, a front, middle, and/or rear port **34** are interchangeable to receive and/or support various accessories **18**. For example, a rear port **34** is located along an axial axis **40** and located at a centerline of the operator's neck. When an accessory **18** is coupled, the rear port **34**, the accessory **18** is aligned with the axial axis **40** to reduce the moment or load on the user's neck.

[0066] FIG. **4** compares a white bill **42** style on the left and a green brim **12** style hard hat **10** in the middle. Bill **42** styled hard hats **10** have a bill **42** in front **14** (e.g., over the user's eyes), but no brim **12** surrounding the sides (e.g., ear) and/or rear **15** (e.g., neck) of hard hat **10**. In contrast, brim **12** styled hard hats **10** have a brim **12** at least partially surrounding and extending about the front **14**, rear **15**, and/or sides of hard hat **10**. For example, FIGS. **1-3** show a brim **12** styled hard hat **10**. Applicant has found that users have preferences for different styles of hard hats **10**. For example, brim **12** styles can be symmetric/balanced from front **14** to rear **15**, whereas bill **42** styles enable a user to rotate earmuffs further upwards without interference from the surrounding brim **12**. Different hard hat **10** styles with different geometries can have the same or similar mounting locations **16** to support the same accessories **18**.

[0067] The composite image of bill **42** and brim **12** hard hats **10** is shown on the left of FIG. **4**. In this composite image, the white bill **42** (e.g., of a bill styled climbing helmet) extends further in the front **14**. The extended brim **12** style (e.g., traditional style) extends further on the sides and rear **15** of hard hat **10**. Both bill **42** and brim **12** hard hats **10** are shown supporting earmuffs **38** in the rearward port **34**. As shown, white bill **42** hard hat **10**, on the left, does not interfere with rotation of earmuffs **38** to a position above and/or behind the user's ear. The middle image shows earmuff **38** stored behind the operator's ear, but under brim **12** to minimize interference with earmuff **38** at the rearward brim **12** of hard hat **10**.

[0068] Often the user attaches various accessories **18** at different locations (e.g., mounting location **16** and/or ports **34**) to maximize utility (e.g., accommodate more accessories **18**) and/or comfort. The composite image on the right shows earmuffs **38** in the rear port **34**. In this position, earmuff **38** extends substantially parallel to axial axis **40** and over the user's ears, when earmuffs **38** are in the shown operating position. For example, a user places earmuff **38** in the third or rear port **34** because this attachment location for earmuff **38** is closer to a centerline, or axial axis **40**, of the operator's neck (e.g., the base of the user's head). Alternative positions for accessories **18** may aid with shade and/or muscle comfort, e.g., by keeping the mass or CG as close to the axial axis **40** at the centerline.

[0069] Stated differently, accessories **18** fit differently on hard hats **10** of different styles and/or geometries. However, where the mounting locations **16** and/or auxiliary ridges **26** are the same, or similar, for different hard hats **10**, they interchangeably receive accessories **18** with a complimentary mounting bracket **24** (or insert).

[0070] FIG. **5** is a cross-sectional comparison of the bill **42** style and the brim **12** style hard hats **10** of FIG. **4**. As shown, in various embodiments, hard hat **10** has a full brim **12** (e.g., brim **12** style) or a large bill **42** without a surrounding brim **12** (e.g., bill **42** style). Mounting locations **16** for the bill **42** and brim **12** style hard hats **10** are different. For example, comparably sized hard hats **10** have an offset between the mounting location **16** on the short brim **12** (brim **12** style) and long bill **42** (bill **42** style) hard hat **10**.

[0071] FIG. **6** shows an extendible visor **48** that surrounds and clamps or directly couples to auxiliary ridges **26** and/or mounting locations **16** on hard hat **10**, e.g., similar to band **32**. Visor **48** couples under mounting locations **16** and/or auxiliary ridges **26** in such a way as to leave ports **34** and mounting locations **16** available for additional accessories **18**. Visor **48** is made from a flexible material, such as a fabric, polymer, or plastic. In one embodiment, visor **48** includes a front portion **50** and a rear portion **52** interconnected by an elastic section **54**. Elastic section **54** enables a user to stretch the visor **48** around the auxiliary ridge **26** and mounting locations **16**.

[0072] Visor **48** couples under mounting locations **16** and auxiliary ridge **26** along an outer perimeter **88** of hard hat **10**. Visor **48** deflects to rotate/move earmuff **38** from below the visor **48** in an operation position **56** to a stored position **58** for earmuff **38** above the visor **48**, as shown in FIG. **6**. In addition, the elastic section **54** enables the visor **48** to provide some slack when earmuff **38** is moved over visor **48** from an operation position **56** to a stored position **58**. The flexible nature of visor **48** and/or elastic section **54** facilitates the movement from use (e.g., operation position **56**) to stored position **58** and back.

[0073] FIG. **7** shows two perspective views of a visor **60** with a mounting bracket **24** that attaches to the mounting locations **16** on hard hat **10**. Visor **60** acts like a visor **48** for specific accessories, e.g., face shield **22**, but can be modular and segmented. For example, visor **60** enhances face shield **22** for different styles and sizes of hard hats **10**. Additional outer mounting locations **16** on visor **60** support accessories **18** on hard hat **10**, and a distending peak or ridge extension **62** protrudes radially outward between the mounting location **16** and an edge of visor **60**. As such, visor **60** provides cross-compatibility between different hard hats **10** of different sizes and styles, including bills **42** and brims **12**. In addition, visor **60** provides additional mounting locations **16** for other accessories **18**.

[0074] Visor **60** provides a space, or bridge, for accessories **18**, such as lamp **20**, to remain attached to hard hat **10** when an operator moves the face shield **22** from an operation position **56** to a stored position **58** and back. Visor **60** also provides a reliable structural abutment, or seal, between the face shield **22** and any style of hard hat **10** with mounting locations **16** (or auxiliary ridges **26**) configured for the bracket **24** on visor **60**. In this way, visor **60** facilitates coupling a welding face shield **22** and a lamp **20** to the mounting location **16**. As a specific example, this configuration is beneficial when welding, since traditional welding face shields **22** are dark to protect the eyes from the arc, but often too dark to see in ambient light. Operators often find it difficult to see and manipulate work-piece objects when the face shield is in the operating position but the welding arc is not illuminated. Lamp **20** can provide sufficient light for the operator to see and manipulate the welding work-pieces before and after the welding process in the absence of the arc.

[0075] Visor **60** provides a structural and spatial bridge between hard hat **10** and face shield **22** for structural support of another attached accessory **18**. This enables two accessories **18** (e.g., a welding mask and a headlamp) to be coupled at the mounting location **16** of hard hat **10**. Ridge extensions **62** on visor **60** (e.g., bracket **24** of visor **60**) provide a structural offset or bridge from hard hat **10** to visor **64** to accessory **18**.

[0076] Visor **60** has a bracket **24** with a duplicate mounting location **16** on an exterior of visor **60** to accommodate the additional mounted accessory **18**. For example, on one side (e.g., an inner surface) of visor **60** is a mounting bracket **24** with ridges **25** similar to lamp **20**. When this bracket **24** of visor **60** couples to mounting location **16**, an opposite, exposed side (e.g., outer surface) of visor **60** forms an available external mounting location **16** to receive and support an additional mounted accessory **18**, such as a lamp **20**, on hard hat **10**.

[0077] To support accessory **18** and/or visor **60**, each mounting location **16** has side ridges **25** on both sides of mounting location **16**. Mounting bracket **24** has a complimentary side receiving slots **30** on both sides of mounting bracket **24**. Side receiving slots **30** are configured to couple to ridges **25** at the mounting locations **16**. For example, the coupling of receiving slots **30** to ridges **25** mounts visor **60** to hard hat **10** and similarly mounts a lamp **20** to visor **60** (FIG. **18**). In this way, visor **60** serves as a bridge between an accessory **18** and the mounting locations **16** of different sizes and styles of hard hats **10**.

[0078] Applicant has found that having a mounting location **16** for the face shield **22** with the same, or similar, ridge extension **62** dimensions ensures the proper abutting visor **64** position for the face shield **22** in the operation position **56** to protect a user's eyes and/or face. Ridge extension **62** structurally interconnects face shield **22** over the brim **12** of hard hat **10**. In one embodiment, ridge extension **62** and face shield **22** abut to form a seal, such as a watertight or hermetic seal. The seal then protects the user from intruding or splashed liquids or debris.

[0079] Different users prefer different bills **42**, brims **12**, and/or hard hat **10** geometries. Different geometries are also advantageous for various worksites and/or jobs. For example, the same user may have one hard hat **10** with a large bill **42** for outdoor worksites and a second hard hat **10** with a small surrounding brim **12** to support a lamp **20** for indoor job sites.

[0080] Applicant has found that using an intermediary modular visor **60** enables the user to attach one face shield **22** that reliably abuts and/or seals against the brim **12** of a variety of hard hats **10**. Visor **60** creates a reliable fitment regardless of the style, size, or geometry of the brim **12** or bill **42** on the hard hat **10**. In this way, an operator with two hard hats **10** can attach, abut, and/or seal the same face shield **22** to visor **60**, and then interchangeably attach and detach the visor **60** and face shield **22** to the operator's preferred hard hat **10**. This enables the operator to customize the hard hat **10** for a particular environment or job. Alternatively, the same user attaches a welding face shield **22** for a welding job and later attaches a different protective eyeglass face shield **22** for woodworking. In other words, visor **60** bridges different sizes and shapes of various hard hats **10** (or face shields **22**) and permits the use of one face shield **22** to fit various sizes of hard hats **10** with complimentary mounting locations **16**.

[0081] In one embodiment, visor **60** includes an oblong locking protrusion **66** in mounting location **16**. An overhang **68** on bracket **24** engages a notch or notch **69** to lock visor **60** on mounting locations **16** of hard hat **10**. The coupled notch **69** and overhang **68** protect against inadvertent bumping or jostling of accessories **18** when visor **60** is coupled to hard hat **10**.

[0082] FIG. **8** shows perspective, top, and bottom views, respectively, of visor **60**. The connection of visor **60** to hard hat **10** creates a ridge extension **62** that covers both hard hats **10** and helmets with bill **42** and brim **12** styles. The top row shows visor **60** attachment to both bill **42** (left climbing helmet) and brim **12** styled (right-traditional brim style) hard hats **10**.

[0083] Visor **60** attaches to both styles and creates a uniform ridge extension **62** with a consistent brim width W. The brim width W extends between the mounting location **16** of hard hat **10** to an edge of the ridge extension **62** on visor **60**. As shown, the brim width W is greater than either bill **42** or brim **12** dimensions. The middle row shows how the visor **60** creates a uniformly sized attachment for an accessory **18** or face shield **22**. The bottom row shows how brim width W is longer than either extension of bill **42** or brim **12** styles. Specifically, the black visor **60** is longer than either the bill **42** (left) or brim **12** (right) of either hard hat **10**.

[0084] FIG. **9** is a top perspective view of an embodiment of a sun-visor or visor **70** with a bracket **24** over hard hat **10**. Visor **70** is the same as or similar to visor **48** and/or visor **64**, except visor **70** includes one or more attachment mechanism, shown as joint **72** and/or a bracket **24**, to securely but removably couple visor **70** to mounting locations **16** or auxiliary ridges **26** of hard hat **10**.

[0085] In a specific embodiment, hard hat system **8** includes hard hat **10** including a shell **11** formed from a rigid material and visor **70** reversibly coupled to hard hat **10**. The shell **11** includes an accessory mounting location **16** located along an outer surface **7** of shell **11**.

[0086] In various embodiments, visor **70** includes a first portion **130** and a second portion **132**. The first portion **130** extends circumferentially around at least a portion of the shell **11** and extends radially outward from the shell **11** forming a sun-blocking flange. The second portion **132** is coupled to the first portion **130** and positioned adjacent to the accessory mounting location **16**. The second portion **132** is deformable relative to the first portion **130** such that the second portion **132** deforms to accommodate and cover a portion of an accessory **18** (e.g., an ear muff) coupled to an accessory mounting location, shown as one of front port **80**, middle port **82**, or rear port **84**. The deformation of second portion **132** allows accessory **18** to extend from the accessory mounting location (e.g., front port **80**, middle port **82**, or rear port **84**) to below the second portion **132**.

[0087] In various embodiments, second portion **132** is formed from a material that is more elastic than a material of the first portion **130**. In various embodiments, second portion **132** includes a first zone **134** on a first side **136** of shell **11**, and a second zone **138** on a second side **140** of shell **11** opposite the first side **136**. In a specific embodiment, the first side **136** is a left side of the shell **11** and the second side **140** is a right side of the shell **11**. In various embodiments, a fastener, shown as joint **72**, couples the visor **70** to the left side of the shell **11** and a second fastener, shown as joint **72**, couples the visor **70** to the right side of the shell **11**. In various embodiments, second portion **132** includes a third zone **142** and a fourth zone **144** separated by a bridge **78**, and the third zone **142** and the fourth zone **144** are both positioned on a first side of the shell **11** (e.g., a left side of the shell **11**, a right side of the shell **11**). The bridge **78** is formed from a material that is less elastic than a material forming the second portion **132**.

[0088] As described in detail below, visor **70** enables a user to fold a brim **12**, e.g., front over the back or back over the front, and retain the fold. Visor **70** couples to mounting locations **16** and/or ports **34** of auxiliary ridge **26** on different styles and sizes of brim **12** and/or bill **42** hard hats **10** (e.g., climbing styled helmets and traditional hard hats **10**). FIG. **10** shows a rear view of hard hat **10** with visor **70** coupled to the rear mounting ridge via bracket **24**, as illustrated in FIG. **9**.

[0089] With reference to FIGS. **9** and **10**, visor **70** includes a front portion, shown as front brim segment **74**, and a rear portion, shown as rear brim segment **76**, that are interconnected by opposite stretch zones **77**. Front brim segment **74** and rear brim segment **76** are similar to front and rear portions **50** and **52** of visor **48**. A fastener, shown as joint **72**, couples two parallel stretch zones **77** to a port **34** on the auxiliary ridge **26** to secure visor **70**. In one embodiment, the front brim segment **74** and/or rear brim segment **76** includes bracket **24** that couples visor **70** to mounting locations **16** (or auxiliary ridges **26**) on hard hat **10**. A non-elastic material or bridge **78** between stretch zones **77** forms integrally with joint **72** (FIG. **9**) or is a separate component that couples to and supports joint **72** (FIG. **10**).

[0090] Front brim segment **74** is moveable between a covering configuration in which the front brim segment **74** extends in a forward direction **71** away from a front **14** of hard hat **10**, and a stowed configuration in which the front brim segment **74** extends in a rearward direction **73** away from a rear **15** of the hard hat **10**. Front brim segment **74** interfaces with the rear brim segment **76** when the front brim segment **74** is in the stowed configuration.

[0091] In various embodiments, clip **24** is coupled to rear brim segment **76** and clip **24** couples visor **70** to hard hat **10** (FIG. **9**). In various embodiments, clip **24** is coupled to front brim segment **74**, and clip **24** couples to rear **15** of hard hat **10** when the front brim segment **74** of visor **70** is in the stowed configuration. In various embodiments, clip **24** couples front brim segment **74** to front **14** of hard hat **10** when front brim segment **74** of visor **70** is in the covering configuration. In various embodiments, clip **24** is coupled to the front brim segment **74** of visor **70**, the clip **24** couples to a front **14** of the hard hat **10** when the visor **70** is in the covering configuration.

[0092] In various embodiments, joint **72** couples to a port **34** in auxiliary ridge **26**. For example, as shown in FIG. **9**, joint **72** couples to a front port **80**. FIG. **10** shows joint **72** coupling visor **70** to a middle port **82**. Similarly, joint **72** can couple to any port **34** (e.g., rear port **84**) of auxiliary ridge **26** and/or a front or rear mounting location **16** on hard hat **10**. Joint **72** location in a front port **80**, middle port **82**, or rear port **84** determines a fold location **86** of visor **70**. Also, the use of a particular port **34** (e.g., front port **80**, middle port **82**, or rear port **84**) facilitates the attachment of other accessories **18** to either the mounting location **16** and/or other ports **34** in auxiliary ridge **26**. For example, using a front port **80** to attach a face shield **22** and/or visor **70** facilitates earmuff **38** attachment at a middle or rear port **82** or **84** of auxiliary ridge **26**.

[0093] In various embodiments, bracket **24** is coupled directly to visor **70** (FIG. **9**) or over visor **70** (FIG. **10**) at the mounting location **16**. FIG. **9** shows bracket **24** coupled to visor **70** and securely coupled to a rear mounting location **16** on hard hat **10**. In contrast, FIG. **10** shows visor **70** wrapped around an outer perimeter **88** of hard hat **10**. Outer perimeter **88** includes the transition perimeter of the front and rear mounting locations **16** and the side auxiliary ridges **26** towards/adjacent the brim **12** of hard hat **10**. Visor **70** has a hollowed elliptical shape (elongated donut) that extends radially from an inner diameter **90** (e.g., adjacent outer perimeter **88** of hard hat **10**) to an outer diameter **92**. Bracket **24** couples to a rear mounting location **16** and captures visor **70** around the outer perimeter **88**, between the hard hat **10** brim **12** and bracket **24** (e.g., under notch **69** but over brim **12** of hard hat **10**). Bracket **24** includes exterior mounting locations **16** and/or a notch **69** to further support and/or couple an additional accessory **18**. An elastic or strap **94** extends under front brim **12** to fold the brims **12** and hold the folded configuration.

[0094] FIGS. **11**A and **11**B compares an axial or parallel stretch visor **70** (FIG. **11**A) with two stretch zones **77** on opposite sides of visor **70** to a radial stretch visor **70** (FIG. **11**B) with only one stretch zone **77**. As shown in FIGS. **11**A and **11**B, the radial stretch visor **70** expands an inner diameter **90** of visor **70** to open a radial circumference of inner diameter **90** and fit visor **70** on different sizes, shapes, and configurations of hard hats **10** and/or accessories **18**. The parallel stretch visor **70** includes stretch zones **77** on opposite sides of visor **70**, e.g., adjacent to auxiliary ridge **26**. The parallel stretch visor **70** enables folding and may include folding locations **86**, such that a user can fold visor **70**. For example, folded visor **70** extends the brim **12** on a front or rear half, while providing unobstructed access to the mounting system (mounting locations **16** and ports **34** on auxiliary ridge **26**) on the other half of hard hat **10**. Parallel stretch visor **70** also enables users to access ports **34** of auxiliary ridges **26** on opposite sides of hard hat **10**.

[0095] FIGS. **12-14** show different views of visor **70** of FIG. **9** in a folded configuration. Fold locations **86** on opposite sides of visor **70** provide pivots to allow the user to flip the front (or rear) brim **12** of visor **70** and attach an accessory to the mounting location **16** on the front of hard hat **10**. Bracket **24** slides over the flipped brim **12** to capture and hold the rotated brim **12** in the folded configuration. Strap **94** is rotated around both brims **12** (e.g., front and rear brim **12**) to hold the brims **12** together in the folded configuration shown. This configuration provides the user access to mounting locations **16** and/or auxiliary ridges **26** without interference from visor **70** and permits the user to modify visor **70** according to the user's preferences and circumstances.

[0096] In various embodiments, visor **70** includes a first folding location **86** on a first side **81** of visor **70**, and a second folding location **87** on a second side **79** of visor **70** opposite the first side **81**. In various embodiments, first folding location **86** includes a stretch zone **77** that is more elastic than neighboring portions of visor **70** (e.g., portions of visor **70** adjacent to stretch zone **77**).

[0097] FIG. **15** shows another embodiment of a modular visor **100**. Modular visor **100** is the same as or similar to visor **48**, **64**, and **70**, except for the differences described. Modular visor **100** includes individual modules or brim segments (e.g., front brim segment **74** and rear brim segment **76**) that couple together around the outer perimeter **88** of hard hat **10**. In this way, modular visor **100** can include different materials, shapes, and/or accessories **18** based on user preference to provide different functionalities. As shown, front brim segment **74** includes a semi-transparent plastic that, for example, blocks UV rays to the user's eyes but permits some visible light through. Joint **72** couples front brim segment **74** to interconnect rear brim segment **76**. Rear brim segment **76** includes two stretch zones **77** and an opaque plastic material brim **12**.

[0098] Strap **94** can be coupled to front brim segment **74** and/or rear brim segment **76**. Joint **72** can be biased or unbiased to provide folding locations **86** for rotation of either modular section. Joints **72** couple front brim segment **74** to rear brim segment **76**. Joints **72** also couple modular visor **100** to auxiliary ridge **26**. In other embodiments, a compressive force generated by stretch zones **77** expanded about outer perimeter **88** retains visor **100**. Various shims **104** are located around outer perimeter **88** to adjust modular visor **100** to the size, shape, and/or style.

[0099] In some embodiments, joint **72** includes mating protrusions **106** and slots **108** between front brim segment **74** and rear brim segment **76**. For example, protrusions **106** couple or snap into slots **108** to couple the front and rear segments **74** and **76** of modular visor **100**. In various embodiments, joint **72** includes other mechanisms for temporarily and securely joining segments of modular visors **100**, such as hook and loop joints, pressure/friction generating joints, metal and/or plastic quick-release joints, fasteners, buckles, and/or adhesives.

[0100] FIG. **16** is a perspective view of modular visor **100** of FIG. **15** coupled to a front **14** and folded visor **70** of FIG. **9** and coupled to a rear **15** of hard hat **10**. Bridge **78** and/or joint **72** couple/form folding locations **86** in opposite parallel stretch zones **77**. For example, joint **72** couples stretch zone **77** to front port **80**, leaving the middle port **82** and rear port **84** available to couple additional accessories **18**. Stated differently, modular visor **100** independently couples to front **14** of hard hat **10**, such as with a clamping frictional force created by a frictional fit or a bracket **24** or inserted fastener. In some embodiments, a single segment (e.g., front brim segment **74**) includes multiple materials, e.g., a first semi-transparent material **110** and a second opaque material **112**. Since the modules or segments of modular visor **100** are independent, front segment **74** and rear segment **76** can also be fabricated from different materials to exhibit different properties and functionality. Alternatively, one segment, e.g., either segment **74** or **76**, can be used in combination with another accessory **18**, such as face shield **22**.

[0101] As illustrated, no physical connection exists between modular visor **100** and folded visor **70**. This configuration enables the functional coupling or pairing of a modular visor **100** and a folded visor **70**. Modular visor **100** is used in combination with other accessories **18** and/or sun-visors (e.g., visors **48**, **67**, and/or visor **70**, or another visor described herein). For example, two or more modular visors **100** (e.g., a front brim segment **74** and a rear brim segment **76**) are combined to extend around the outer perimeter **88** of hard hat **10** entirely. Front brim segment **74** and rear brim segment **76** of modular visor **100** can be different materials and/or have different transparencies (e.g., transparent, semi-transparent, or opaque). In addition, front brim segment **74** and/or rear brim segment **76** are interchangeably removed and replaced with one or more accessories **18** to fit the operator's preferences, safety, and/or use at the specific job site.

[0102] FIG. **17** is a top perspective view of modular visor **100** coupled to face shield **22** accessory **18**. Face shield **22** includes bracket **24** that couples to front **14** mounting locations **16** on hard hat **10**. Face shield **22** extends around the user's face. In various embodiments, face shield **22** can create a seal between the face (e.g., eyes, nose, and/or mouth) of a user, An optional guard **114** (e.g., a visor sized for face shield **22**) extends outward to form a scam **115** along a joint between face shield **22** and guard **114**. In various embodiments, seam **115** is hermetic, watertight, or another secured seal to prevent debris (e.g., hot/melted metal in a welder helmet) from entering around the user's face. Bracket **24** on face shield **22** includes outer mounting locations **116** with side ridges **125** and a notch **169** that is the same as or similar to the geometry of the mounting locations **16**. Thus, the mounting system enables the use of additional mounting location **116** to attach another accessories **18** (e.g., a lamp and/or eyeglass holder).

[0103] FIG. **18** is a top perspective view of a front modular visor **100** made from a semi-transparent material coupled to a rear modular visor **100** made from an opaque material. Front and rear segments **74** and **76** are coupled to form a modular visor **100** that is coupled to hard hat **10** with a joint **72**. Joint **72** includes protrusions **106** coupled to slots **108** to adjust the side of inner diameter **90** of the combined modular visor **100** (e.g., the visor resulting from the joining of front and rear modular visors **100**). This feature enables a user to modify the size of inner diameter **90** to accommodate the size, style, and/or dimensions of mounting locations **16** and/or auxiliary ridges **26**. As shown in FIG. **18**, outer diameter **92** can increase from the solid line to the dotted outer diameter **92** based on changes to the inner diameter **90**. Front brim segment **74** is made from a different material than rear brim segment **76** to form a multi-material modular visor **100**. For example, front brim segment **74** includes a semi-transparent tinted plastic material. Rear brim segment **76** includes an opaque fabric material coupled to a polymer center. Modular visor **100** has front and rear brim segments **74** and **76** and includes different material features.

[0104] In some embodiments, modular visor **100** includes one or more materials and/or stretch zones **77** to adjust the frictional clamping force of inner diameter **90** of visor **100** on outer diameter **92** of hard hat **10**. In various embodiments, the frictional clamping force and/or bracket **24** couple visor **100** to hard hat **10**. Bracket **24** either forms a part of brim segment **74** or **76** (e.g., is affixed to) or fits over and captures brim segment **74** and/or **76** against hard hat **10** (e.g., is not joined but proximate to segments **74** and **76**).

[0105] FIG. **19** shows another embodiment visor **150** with an enclosed clastic area **152** in the brim **12**. Visor **150** is the same as or similar to other visors described herein (e.g., visors **48**, **64**, **70**, and/or modular visor **100**, but combines different features and modifications from these visors. For example, enclosed elastic area **152** of visor **150** extends from inner diameter **90** of brim **12** but does not extend across brim **12** to outer diameter **92**. Unlike stretch zones **77** of visor **70** the clastic area **152** is enclosed within brim **12** and adjacent to auxiliary ridge **26**. Enclosed elastic area **152** is surrounded by a non-elastic region **154** of brim **12** on visor **150**. Thus, the outer diameter **192** is firm (e.g., not extendible or flexible), but deflections near the sides of hard hat **10** (e.g., near auxiliary ridges **26**) provide access to ports **34** in auxiliary ridge **26**.

[0106] Visor **150** also couples rear brim segment **76** to both front ports **80** and rear mounting location **16**. This configuration creates a stable visor **150** connection and also enables the operator to insert and attach earmuffs **38** to auxiliary ridge **26** through elastic area **152**. Specifically, the operator passes a four-bar support arm **156** linkage for either face shield **22** or earmuffs **38**. Support arm **156** passes into or through elastic area **152** and couples inserts **158** (at the terminus of support arm **156**) into the remaining middle or rear port **82** or **84** of each opposite auxiliary ridge **26** (FIG. **20**). Coupling rear brim segment **76** to front port **80** and rear mounting location **16** improves the ergonomic attachment of earmuffs **38** to either a middle port **82** or rear port **84** that aligns most comfortably over the individual user's ear.

[0107] FIG. **20** shows different views of the enclosed elastic area visor **150** and support arm **156** to insert **158** and couple earmuff **38** within the rear port **84**. As described above, some users may opt to use the middle port **82** to fit earmuff **38** directly over their ear ergonomically. In contrast, the rear port **84** may be a better fit for other users. The modular configuration of visor **150** coupled to front port **80** enables the user to optionally choose the configuration that is most ergonomic and comfortable. This configuration also keeps another port **34** available on each side of auxiliary ridge **26** for another accessory **18** (e.g., reflector, eyeglass holder, or tool carrier).

[0108] FIG. **21** shows another embodiment of visor **160** with mixed materials. Visor **160** includes different materials with a rigid outer perimeter or outer diameter **92** and a soft fabric inner diameter **90** to accommodate differences in the sizes, styles, and/or dimensions of various hard hats **10** or helmets with different sized mounting locations **16** or auxiliary ridges **26**. For example, visor **160** includes two stretch zones **77** interconnecting front and rear brims **12**. Brim **12** include a pad **162** and bracket **24** to couple visor **160** to different sizes and shapes of hard hats **10** and/or attached accessories **18** (e.g., face shield **22**).

[0109] In various embodiments, face shield **22** can be a convertible face shield **22** that attaches to hard hat **10** when visor **100** is folded towards the rear **15** of hard hat **10** (FIG. **17**). Alternatively, face shield **22** couples to a rear segment **74** of modular visor **150**. Visor **160** combines various features of visors previously described herein (e.g., visors **48**, **64**, **70**, **100**, and/or **150**) to further customize attachment of different accessories **18** (e.g., face shield **22**) to mounting locations **16** and/or auxiliary ridges **26**.

[0110] Visor **160** mounts around hard hat **10** perimeter **88** with a coupled accessory **18** (face shield **22**) at a front mounting location **16** of hard hat **10**. This configuration demonstrates how the features and/or materials of visor **160** interact with various attached accessories **18** (e.g., four-bar linkages, support arms, and/or inserts) at mounting locations **16** and/or auxiliary ridges **26**. FIG. **22** is an isolated perspective view of visor **160** of FIG. **21**.

[0111] As illustrated in FIGS. **21-22**, bracket **24** includes an external mounting location **116** that is the same as or similar to the mounting location **16** on hard hat **10**. In some embodiments, bracket **24** includes external mounting locations **116** with ridges **125**. In this way, bracket **24** serves as a bridge that couples a first accessory **18** (e.g., face shield **22**) to front mounting location **16** of hard hat **10** and provides a second outer mounting location **116** to attach a second accessory **18**. Stated differently, bracket **24** forms a bridge of connections that enables stacking multiple accessories **18** onto a single mounting location **16** of hard hat **10**.

[0112] A front pad **162** along the inner diameter **90** of a front transparent brim segment **74** provides a tolerance **165** to fit visor **160** on different sizes and shapes of various accessories, hard hats **10**, and helmets. Two opposite and parallel stretch zones **77** are located adjacent auxiliary ridge **26** and extend from inner diameter **90** to outer diameter **92** of brim **12**. A spring-loaded or biased fastener creates a biased pivot **164** (e.g., similar to joint **72**) that couples stretch zones **77** of visor **160** to a port **34** on auxiliary ridge **26** (e.g., front port **80**, middle port **82**, or rear port **84**). Biased pivot **164** creates the bias or spring force against visor **160**, such that the operator applies a downward force on visor **160** to create a clamping frictional force at the inner diameter **92** on brim **12**. Stretch zones **77** include a stretched flexible material that extends from inner diameter **90** to outer diameter **92** to create a compressive force on outer perimeter **88** and enable pad **162** and visor **160** to deflect for different sizes, shapes, and styles of hard hat **10**. A wire **166** extends along outer diameter **92** of visor **160** to create a rigid outer diameter **92**.

[0113] An optional bracket **24** couples to a rear segment **76** of visor **160**. Bracket **24** couples or captures visor **160**. Pad **162** slides under a notch **69** at mounting location **16** of hard hat **10**. As illustrated, mounting location **16** is coupled to face shield **22**, and pad **162** fits under notch **169** to couple to the outer perimeter **88** of hard hat **10** with the attached accessory **18**. Stretch zones **77** are elastic/flexible adjacent auxiliary rides **26**, but wire **166** forms a rigid/firm outer diameter **92** of visor **160**. This configuration enables multiple materials in semi-transparent or tinted front and rear segments **74** and **76**, stretch zones **77**, and pad **162** to form expandable features at the joint between outer perimeter **88** of hard hat **10** and inner diameter **90** of visor **160**. In contrast, outer diameter **92** is relatively rigid and firm through the various materials due to wire **166**. In one embodiment, stretch zones **77** extend over and/or deflect around both support arms **156** and inserts **158** that attach face shield **22** and/or earmuffs **38**.

[0114] Referring to FIGS. **23-27**, various aspects of a visor **220** are shown. Visor **220** is substantially the same as visor **48**, visor **60**, visor **64**, visor **67**, visor **70**, visor **100**, or visor **150**, except for the differences discussed herein. Visor **220** includes a rear portion **224** coupled to hard hat **210** via frame **240**. Side portions **222** extend from rear portion **224** circumferentially extend around hard hat **210**. Folding portion **232** is coupled to rear portion **224** via folding location **230** at connecting portion **236**. In a specific embodiment, rear portion **224** and folding portion **232** are formed from a fabric or fabric-like material.

[0115] Frame **240** is coupled to safety headwear, shown as hard hat **210**, at one of a plurality of mounting slots along the side of hard hat **210**. In the specific embodiment shown, frame **240** is coupled to a middle port **214** of hard hat **210**. In a specific embodiment, frame **240** is formed in an arch, U or horseshoe-shape and is formed from a plastic material.

[0116] Folding portion **232** includes a frontal relief portion, shown as front flap **226**, configured to receive various accessories, such as the arms of ear muffs. Similarly, side portion **222** includes a rear relief portion, shown as rear flap **228**, configured to receive various accessories, such as the arms of ear muffs. In use, front portion **234** and folding portion **232** can be folded up and over the dome of hard hat **210** and backwards towards rear portion **224** (see FIG. **27**). In this position, front portion **234** and/or folding portion **232** can be coupled to a coupling element, shown as clip **242** (FIG. **24**), to hold these portions of visor **220** in the folded position.

[0117] In a specific embodiment, visor **220** includes first portion **221** and second portion **223**. Similar to various embodiments of visor **70**, second portion **223** is more deformable than the first portion **221** such that the second portion **223** deforms to accommodate and cover a portion of an accessory (e.g., an arm of an ear muff) coupled to an accessory mounting location below second portion **223**. In a various embodiments, second portion **223** is less taut than first portion **221**. In various embodiments, second portion **223** includes front flap **226** and rear flap **228**. For example, second portion **223** is more deformable without being stretched compared to how deformable first portion **221** is without being stretched (e.g., because second portion has extra fabric that is not as taut). As another example, second portion **223** extends upwardly and loosely upwards against an exterior of the shell (see FIG. **26**). In a specific embodiment, rear portion **224** includes a relief portion, shown as rear flap **228**, that is less taut than portions of the visor **220** adjacent the rear flap **228**, and rear flap **228** is positioned below the front portion **234** and/or the folding portion **232** when the visor **220** is in the stowed configuration.

[0118] Referring further to FIGS. **25-27**, when visor **220** is not folded (FIG. **26**), an accessory, such as arms of an ear muff, extend through rear flap **228**. When visor **220** is folded (FIG. **27**), an accessory, such as arms of an ear muff, extend through front flap **226** and rear flap **228**. In a specific embodiment, visor **220** includes one or more flexible portions, shown as flex points **238**, that permit visor **220** to form various shapes to accommodate various accessories being covered, surrounding or transiting beneath front flap **226** and/or rear flap **228**. In various embodiments, a front portion of visor **220** includes a front flap **226** that is less taut than portions of the visor **220** adjacent the front flap **226**, and the rear portion includes a rear flap **228** that is less taut than portions of the visor **220** adjacent the rear flap **228**. In a specific embodiment, the front flap **226** is positioned above the rear flap **228** when the visor **220** is in the stowed configuration (e.g., when the front portion of visor **220** is folded backward, such as shown in FIG. **12**).

[0119] Referring to FIGS. **28-30**, various aspects of visor **220** being coupled to different safety headwear are shown. Frame **240** is coupled higher to some safety headwear, shown as helmet **212** in FIG. **28**, than other safety headwear, shown as hard hat **210** in FIG. **29**. If unaddressed, this results in frame **240** and portions of visor **220** extending from helmet **212** at a steeper angle compared to frame **240** and visor **220** coupled to hard hat **210**.

[0120] To compensate for this, frame **240** includes clip **244**. Clip **244** includes an engaging portion **258** that includes a first surface, such as an upper surface, shown as upper step surface **256**, and a second surface, such as a lower surface, shown as lower step surface **254**, each of upper step surface **256** and lower step surface **254** facing in an upward direction **257**. In use, when coupling clip **244** to helmet **212**, upper step surface **256** or lower step surface **254** is coupled to helmet **212**. In particular, clip **244** is coupled to hard hat **210** in a first position or in a second position. The upper step surface **256** engages with the hard hat **210** when the clip **244** is in the first position and the lower step surface **254** engages with the hard hat **210** when the clip **244** is in the second position. In a specific embodiment, the upper step surface **256** is spaced a distance **253** above the lower step surface **254**. Therefore, clip **244** is inserted lower into the body of helmet **212** compared to hard hat **210**, and as a result the angle of frame **240** and visor **220** provides more consistency when coupled to different embodiments of safety headwear. In a specific embodiment, engaging portion **258** is biased in a horizontal direction against hard hat **210**.

[0121] Referring to FIGS. **28-29**, the visor **220** extends from the hard hat **210** at a first angle **247** relative to horizontal when the clip **244** is coupled to the hard hat **210** in the first position, and the visor **220** extends from the hard hat **210** at a second angle **248** relative to horizontal when the clip **244** is coupled to the hard hat **210** in the second position. In a specific embodiment, first angle **247** is different than the second angle **248**, and more specifically first angle **247** is greater than second angle **248**. In various embodiments, angle **247** and angle **248** are measured from a front of hard hat **210**. In various embodiments, lower step surface **254** is spaced a second distance **259** closer to a center **218** of the hard hat **210** than the upper step surface **256**.

[0122] In various embodiments, hard hat system **208** includes hard hat **210**, clip **244** coupled to hard hat **210**, and visor **220** coupled to the visor **220** such that the visor **220** is reversibly supported to the hard hat **210** via the clip **244** and extends circumferentially around a portion of the hard hat **210**.

[0123] Referring to FIGS. **31-32**, various aspects of visor **220** are shown, such as enabling multiple objects to be coupled to the same general area of helmet **212**. In a specific embodiment, frame **240** includes a coupling element, shown as rear slot **250**. Rear slot **250** is configured to duplicate the coupling capabilities of the rear portion of helmet **212**, thereby allowing a user to "daisy chain" elements to the same general area of helmet **212**. For example, a user may couple frame **240** to helmet **212** and use rear slot **250** to couple additional accessories, such as headlamp battery **252**.

[0124] Referring to FIGS. **33-35**, in various embodiments, visor **220** includes components to provide flexibility to couple to safety headwear of different sizes. In such embodiments, a front portion **234** of visor **220** is coupled to brim **216** of helmet **212** via clip **246**. In a specific embodiment, clip **246** is formed from a plastic material.

[0125] In a specific embodiment, connecting portions **236** elastically couples front portion **234** and rear portion **224**. The elasticity of connecting portions **236** permits visor **220** to be securely coupled to various sized embodiments of safety headwear (compare FIG. **33** and FIG. **35**).

[0126] Referring to FIG. **36**, various aspects of a coupling element, shown as spring clip **262**, are shown. Spring clip **262** helps retain the visor on the rear retention feature of safety headwear. Spring clip **262** includes an interfacing element, shown as tab **264**. A user can interface with tab **264** to pull in direction **260** and thereby disengage spring clip **262**.

[0127] Referring to FIG. **37**, various aspects of a safety headwear, shown as hard hat **310**, are shown. Hard hat **310** is similar to hard hat **210** except for the differences described herein. Hard hat **310** includes a first coupling feature at a front, shown as front mount **312**, and a second coupling feature, shown as rear mount **316**, at a rear of hard hat **310** opposite the front. Hard hat **310** includes a coupling element **314** at a front of hard hat **310**. Hard hat **310** includes a centrally-located coupling element on a side of hard hat **310**, shown as slot **322**, located between rear slot **320** and front slot **324**. It is contemplated herein that hard hat **310** may interface with one or more elements described herein, such as for exemplary purposes only, visor **220**.

[0128] Referring to FIGS. **38-39**, various aspects of frame **420** are shown. Frame **420** is substantially the same as frame **240** except for the differences discussed herein.

[0129] Frame **420** couples to safety headwear, such as hard hats, via coupling mechanisms, shown as hooks **422** (e.g., hooks **422** engage with ridges **25** of mounting location **16** of hard hat **10**, see FIG. **3**). One or more of tabs **448** interface with a protrusion extending from a front of the hard hat, such as below the protrusion.

[0130] Frame **420** includes one or more coupling elements, shown as engaging portions **460**. Engaging portions **460** are substantially the same as engaging portions **258** except for the differences discussed herein. In particular, engaging portions **460** include an extension that extends vertically above the body of engaging portions **460**.

[0131] In use, hooks **422** are slideably engaged with the hard hat to restrict frame **420** to vertical movement relative to the hard hat, and tabs **448** bias frame **420** from moving upward and disengaging from the hard hat. To remove frame **420** from the hard hat, a user pulls lever **440** in upward direction **442**. In response, pivot **446** interfaces against the hard hat, which forces tabs **448** away from the hard hat until tabs **448** do not interface with the protrusion extending from the hard hat. As a result, the user can slide frame **420** upward relative to the hard hat to disengage frame **420** from the hard hat.

[0132] Referring to FIG. **40**, various aspects of hard hat system **508** are shown. Hard hat system **508** includes face shield **510** and frame **520**. Hard hat system **508** is substantially the same as hard hat system **8** except for the differences discussed herein, and frame **520** is substantially the same as frame **420** except for the differences discussed herein. In particular, hooks **522** of frame **520** extend circumferentially along an interior of frame **420** further than hooks **422** of frame **420**.

[0133] Referring to FIG. **41**, various aspects of frame **620** are shown. Frame **620** is substantially the same as frame **420** or frame **520** except for the differences discussed herein. In particular, frame **620** includes structure **624** above hooks **622** that extends outward from the hard hat that frame **620** is coupled to.

[0134] It should be understood that the figures illustrate the exemplary embodiments in detail, and it should be understood that the present application is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology is for the purpose of description only and should not be regarded as limiting.

[0135] Further modifications and alternative embodiments of various aspects of the invention will be apparent to those skilled in the art in view of this description. Accordingly, this description is to be construed as illustrative only. The construction and arrangements, shown in the various exemplary embodiments, are illustrative only. Although only a few embodiments have been described in detail in this disclosure, many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter described herein. Some elements shown as integrally formed may be constructed of multiple parts or elements, the position of elements may be reversed or otherwise varied, and the nature or number of discrete elements or positions may be altered or varied. The order or sequence of any process, logical algorithm, or method steps may be varied or re-sequenced according to alternative embodiments. Other substitutions, modifications, changes, and omissions may also be made in the design, operating conditions, and arrangement of the various exemplary embodiments without departing from the scope of the present invention.

[0136] For purposes of this disclosure, the term "coupled" means the joining of two components directly or indirectly to one another. Such joining may be stationary in nature or movable in nature. Such joining may be achieved with the two members and any additional intermediate members being integrally formed as a single unitary body with one another or with the two members or the two members and any additional member being attached to one another. Such joining may be permanent in nature or alternatively may be removable or releasable in nature.

[0137] While the current application recites particular combinations of features in the claims appended hereto, various embodiments of the invention relate to any combination of any of the features described herein whether or not such combination is currently claimed, and any such combination of features may be claimed in this or future applications. Any of the features, elements, or components of any of the exemplary embodiments discussed above may be used alone or in combination with any of the features, elements, or components of any of the other embodiments discussed above.

[0138] In various exemplary embodiments, the relative dimensions, including angles, lengths, and radii, as shown in the Figures, are to scale. Actual measurements of the Figures will disclose relative dimensions, angles, and proportions of the various exemplary embodiments. Various exemplary embodiments extend to various ranges around the absolute and relative dimensions, angles, and proportions that may be determined from the Figures. Various exemplary embodiments include any combination of one or more relative dimensions or angles that may be determined from the Figures. Further, actual dimensions not expressly set out in this description can be determined by using the ratios of dimensions measured in the Figures in combination with the express dimensions set out in this description. In addition, in various embodiments, the present disclosure extends to a variety of ranges (e.g., plus or minus 30%, 20%, or 10%) around any of the absolute or relative dimensions disclosed herein or determinable from the Figures.

What is claimed is:

1. A hard hat system comprising:
a hard hat comprising:
a shell; and
an accessory mounting ridge positioned on an exterior surface of the shell;
a frame supported by the hard hat;
a visor coupled to the hard hat via the frame, the visor comprising:
a first portion extending circumferentially around a rear portion of the shell and extending radially outward from the shell;
a second portion coupled to the first portion and positioned adjacent to the accessory mounting ridge; and
a third portion extending from the second portion circumferentially around a front portion of the shell and extending radially outward from the shell.

2. The hard hat system of claim **1**, wherein the second portion of the visor includes a flap, and wherein the flap is configured to receive an accessory.

3. The hard hat system of claim **1**, the second portion comprising a first section on a first lateral side of the shell and a second section on a second lateral side of the shell opposite the first lateral side.

4. The hard hat system of claim **3**, wherein the frame extends around the rear portion of the shell, and wherein the rear portion of the shell is positioned between the first lateral side of the shell and the second lateral side of the shell.

5. The hard hat system of claim **1**, wherein the accessory mounting ridge of the hard hat further comprises a port and wherein the frame is coupled to the port.

6. The hard hat system of claim **5**, wherein the frame further comprises a clip and wherein the clip is positioned within the port.

7. The hard hat system of claim **1**, wherein the frame has a U-shape.

8. A hard hat system comprising:
a hard hat comprising:
a shell; and
an auxiliary mounting ridge located on a lateral, exterior surface of the shell;
a frame supported by the hard hat;
a visor coupled to the hard hat via the frame, the visor comprising:
a first portion extending circumferentially around a portion of the shell and extending radially outward from the shell; and
a second portion coupled to the first portion and positioned adjacent to the auxiliary mounting ridge, the second portion comprising a first relief portion; and
a third portion extending from the second portion circumferentially around another portion of the shell and extending radially outward from the shell.

9. The hard hat system of claim **8**, wherein the first relief portion is a flap.

10. The hard hat system of claim **8**, wherein the first relief portion is a first flap and wherein the second portion of the visor further comprises a second flap.

11. The hard hat system of claim **10**, wherein the first flap is positioned between the third portion of the visor and the second flap.

12. The hard hat system of claim **8**, wherein the first relief portion is positioned along an interior edge of the visor and adjacent to the hard hat.

13. The hard hat system of claim **8**, the visor comprising a first folding location on a first side of the visor and a second folding location on a second side of the visor opposite the first side.

14. The hard hat system of claim **8**, wherein the visor further comprises a flex point positioned between the first relief portion and the first portion of the visor.

15. A hard hat system comprising:
a hard hat comprising:
a shell; and
an auxiliary mounting ridge located on an exterior surface of the shell;
a visor coupled to the hard hat, the visor comprising:
a brim extending circumferentially around the exterior surface of the shell and radially outward from the exterior surface; and
an enclosed elastic area within the brim, the enclosed elastic area positioned adjacent to the auxiliary mounting ridge.

16. The hard hat system of claim **15**, wherein the enclosed elastic area extends outward from an inner diameter of the brim.

17. The hard hat system of claim **16**, wherein an outermost portion of the enclosed elastic area is spaced from an outer diameter of the brim.

18. The hard hat system of claim **15**, wherein the brim comprises a non-elastic region that surrounds the enclosed elastic area such that an outer diameter of the brim is rigid.

19. The hard hat system of claim **15**, wherein the brim is couplable to a port of the auxiliary mounting ridge.

20. The hard hat system of claim **15**. wherein the enclosed clastic area is configured to allow insertion of an accessory through the brim for coupling to the auxiliary mounting ridge.

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